

P25 J3362: Non-uniform Refinement

CryoSPARC Running Version: v4.7.1

Project Details

DETAIL	VALUE
Title	24sep20b_WT-Control-White_From-January
Description	This is a repeat collection of White Control Wildtype CFTR from the original January/February 2024 collection. Using
ID	P25
Created by user	Andy Paige

Job Details

DETAIL	VALUE
Title	New Job J3362
Description	Enter a description.
ID	J3362
Job type	Non-uniform Refinement
Status	completed
Created by user	Minou Emmad
Created	Fri Jun 05 2026 17:41:02 GMT-0400 (Eastern Daylight Time)
Queued	Fri Jun 05 2026 17:41:38 GMT-0400 (Eastern Daylight Time)
Launched	Fri Jun 05 2026 17:51:28 GMT-0400 (Eastern Daylight Time)
Started	Fri Jun 05 2026 17:51:36 GMT-0400 (Eastern Daylight Time)
Waiting	-
Killed	-
Completed	Fri Jun 05 2026 18:20:36 GMT-0400 (Eastern Daylight Time)
Failed	-
Last accessed	Wed Jun 10 2026 15:51:03 GMT-0400 (Eastern Daylight Time)
Size	6.22 GB

Inputs

particles

J3356.imported_particles

BLOB
J3356.imported_particles.blob.F

CTF
J3356.imported_particles.ctf.F

ALIGNMENTS3D

```
+-----+
| J3356.imported_particles.alignments3D.F |
+-----+
```

volume

```
J3359.volume_class_0
+-----+
| MAP |
+-----+
| J3359.volume_class_0.map.F |
+-----+
```

mask

No group connections

Parameters

Particle preprocessing

PARAMETER	VALUE	DEFAULT	SPEC	ADVANCED
Window dataset (real-space)	True	X		
Window inner radius	0.85	X		X
Window outer radius	0.99	X		X

Homogeneous Refinement

PARAMETER	VALUE	DEFAULT	SPEC	ADVANCED
Symmetry	C1	X		
Symmetry relaxation method	none	X		X
Do symmetry alignment	True	X		X
Re-estimate greyscale level of input reference	True	X		X
Number of extra final passes	0	X		X
Maximum align resolution (A)	None	X		
Initial lowpass resolution (A)	30	X		X
GSFSC split resolution (A)	20	X		X
Force re-do GS split	False	X		X
Adaptive Marginalization	True	X		X
Non-uniform refine enable	True	X		X
Non-uniform filter order	8	X		X
Non-uniform AWF	3	X		X
Enforce non-negativity	False	X		X
Window structure in real space	True	X		X
Skip interpolant premult	True	X		X
Ignore DC component	True	X		X
Ignore tilt	False	X		X
Ignore trefoil	False	X		X
Ignore tetra	False	X		X
Use random ordering of particles	True	X		X

	Disable auto batchsize		False		X				X	
	Batchsize epsilon		0.001		X				X	
	Batchsize snrfactor		50		X				X	
	Reset input per-particle scale		True		X				X	
	Minimize over per-particle scale		False		X				X	
	Scale min/use start iter		0		X				X	
	Noise priorw		50		X				X	
	Noise initw		200		X				X	
	Noise initial sigma-scale		3		X				X	
	Initialize noise model from images		False		X				X	
	Dynamic mask threshold (0-1)		0.2		X				X	
	Dynamic mask near (A)		6		X				X	
	Dynamic mask far (A)		14		X				X	
	Dynamic mask start resolution (A)		12		X					
	Dynamic mask use absolute value		False		X				X	
	Show plots from intermediate steps		True		X					
	GPU batch size of images		None		X				X	

Defocus Refinement

	PARAMETER		VALUE		DEFAULT		SPEC		ADVANCED	
	Optimize per-particle defocus		False		X				X	
	Num. particles to plot		3		X					
	Minimum Fit Res (A)		20		X					
	Defocus Search Range (A +/-)		2000		X					
	GPU batch size of images		None		X				X	

Global CTF Refinement

	PARAMETER		VALUE		DEFAULT		SPEC		ADVANCED	
	Optimize per-group CTF params		False		X				X	
	Num. groups to plot		3		X					
	Binning to apply to plots		1		X				X	
	Minimum Fit Res (A)		10		X					
	Fit Tilt		True		X					
	Fit Trefoil		True		X					
	Fit Spherical Aberration		False		X					
	Fit Tetrafoil		False		X					
	Fit Anisotropic Mag.		False		X					

GPU batch size of images	None	X		X	

Ewald Sphere Correction

PARAMETER	VALUE	DEFAULT	SPEC	ADVANCED	
Do EWS correction	False	X		X	
EWS curvature sign	negative	X		X	

Random Seeds

PARAMETER	VALUE	DEFAULT	SPEC	ADVANCED	
Random seed	991760986	X		X	

Compute settings

PARAMETER	VALUE	DEFAULT	SPEC	ADVANCED	
Cache particle images on SSD	True	X			
Low-Memory Mode	False	X			

Outputs

particles

ALIGNMENTS3D	
J3362.particle.alignments3D	

CTF	
J3362.particle.ctf	

BLOB (PASSTHROUGH)	
J3362.particle.blob	

volume

MAP	
J3362.volume.blob	

MAP_SHARP	
J3362.volume.blob	

MAP_HALF_A	
J3362.volume.blob	

MAP_HALF_B	
J3362.volume.blob	

MASK_REFINE	

J3362.volume.blob	
MASK_FSC	
J3362.volume.blob	
MASK_FSC_AUTO	
J3362.volume.blob	
PRECISION	
J3362.volume.blob	

mask

MASK_REFINE	
J3362.volume.blob	

Target

ganges.biology.columbia.edu (node)

```
> [2026-06-05 17:51:28.24 GMT-4] License is valid.
> [2026-06-05 17:51:28.24 GMT-4] Launching job on lane default target ganges.biology.columbia.edu ...
> [2026-06-05 17:51:28.32 GMT-4] Running job on master node hostname ganges.biology.columbia.edu
> [2026-06-05 17:51:36.69 GMT-4] [CPU: 78.9 MB]      [Avail: 297.05 GB] Job J3362 Started
> [2026-06-05 17:51:36.75 GMT-4] [CPU: 78.9 MB]      [Avail: 296.96 GB] Master running v4.7.1, worker running v4.7.1
> [2026-06-05 17:51:36.77 GMT-4] [CPU: 79.2 MB]      [Avail: 296.96 GB] Working in directory:
                                /mnt/nfs_HuntLabSynologyDS1621CryoEM/hCFTR/Movies/24sep20b/24
                                sep20b/rawdata/CS-24sep20b-wt-control-white-from-january/J336
                                2
> [2026-06-05 17:51:36.78 GMT-4] [CPU: 79.2 MB]      [Avail: 297.00 GB] Running on lane default
> [2026-06-05 17:51:36.78 GMT-4] [CPU: 79.2 MB]      [Avail: 296.99 GB] Resources allocated:
> [2026-06-05 17:51:36.78 GMT-4] [CPU: 79.2 MB]      [Avail: 296.99 GB]   Worker:  ganges.biology.columbia.edu
> [2026-06-05 17:51:36.79 GMT-4] [CPU: 79.2 MB]      [Avail: 297.02 GB]   CPU   :  [0, 1, 6, 7]
> [2026-06-05 17:51:36.79 GMT-4] [CPU: 79.2 MB]      [Avail: 297.06 GB]   GPU   :  [0]
> [2026-06-05 17:51:36.79 GMT-4] [CPU: 79.2 MB]      [Avail: 297.05 GB]   RAM   :  [0, 3, 4]
> [2026-06-05 17:51:36.80 GMT-4] [CPU: 79.2 MB]      [Avail: 297.04 GB]   SSD   :  True
> [2026-06-05 17:51:36.80 GMT-4] [CPU: 79.2 MB]      [Avail: 297.04 GB] -----
                                -
> [2026-06-05 17:51:36.80 GMT-4] [CPU: 79.2 MB]      [Avail: 297.03 GB] Importing job module for job type nonuniform_refine_new...
> [2026-06-05 17:51:48.57 GMT-4] [CPU: 273.7 MB]     [Avail: 301.62 GB] Job ready to run
> [2026-06-05 17:51:48.58 GMT-4] [CPU: 273.7 MB]     [Avail: 301.62 GB] *****
                                **
> [2026-06-05 17:51:49.49 GMT-4] [CPU: 337.4 MB]     [Avail: 301.06 GB] Using random seed of 991760986
> [2026-06-05 17:51:49.52 GMT-4] [CPU: 343.2 MB]     [Avail: 301.06 GB] Loading a ParticleStack with 65008 items...
> [2026-06-05 17:51:49.55 GMT-4] [CPU: 343.2 MB]     [Avail: 301.06 GB]

                                SSD cache ACTIVE at
                                /scr/instance_ganges.biology.columbia.edu:39001 (200 GB
                                reserve) (3 TB quota)

                                Cache usage                                Amount

                                Total / Usable                            3.58 TiB / 2.73 TiB
                                Used / Free                              2.73 TiB / 58.20 MiB
                                Hits / Misses                            90.26 GiB / 0.00 B
                                Acquired / Required                      90.26 GiB / 0.00 B

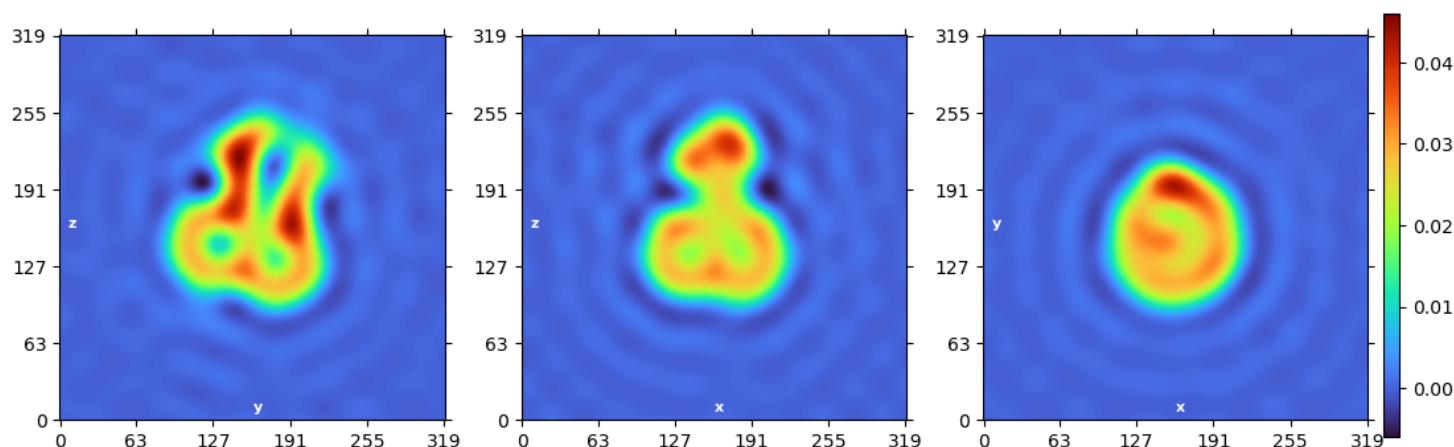
                                Progress:
                                [
                                10994/10994 (100%)
                                Elapsed: 0h 00m 10s
                                Active jobs: P25-J3361, P31-J189, P25-J3362
                                SSD cache complete for 10994 file(s)

> [2026-06-05 17:52:02.59 GMT-4] [CPU: 411.9 MB]     [Avail: 300.79 GB] Done.
> [2026-06-05 17:52:02.60 GMT-4] [CPU: 411.9 MB]     [Avail: 300.79 GB] Windowing particles
> [2026-06-05 17:52:02.61 GMT-4] [CPU: 412.0 MB]     [Avail: 300.79 GB] Done.
> [2026-06-05 17:52:02.61 GMT-4] [CPU: 412.0 MB]     [Avail: 300.79 GB] ===== Gold Standard Split =====
> [2026-06-05 17:52:02.61 GMT-4] [CPU: 412.0 MB]     [Avail: 300.79 GB] Particles have input alignments3D connected, so reusing
                                pre-existing split
> [2026-06-05 17:52:02.61 GMT-4] [CPU: 412.1 MB]     [Avail: 300.79 GB] Set A is smaller than set B by 440 particles (0.677 percent
                                difference relative to the total dataset).
> [2026-06-05 17:52:02.87 GMT-4] [CPU: 426.9 MB]     [Avail: 301.03 GB] Split A has 32284 particles
> [2026-06-05 17:52:02.87 GMT-4] [CPU: 426.9 MB]     [Avail: 301.03 GB] Split B has 32724 particles
> [2026-06-05 17:52:02.88 GMT-4] [CPU: 426.9 MB]     [Avail: 301.04 GB] ===== Refinement =====
> [2026-06-05 17:52:02.88 GMT-4] [CPU: 426.9 MB]     [Avail: 301.04 GB] Input particles have box size 320
> [2026-06-05 17:52:02.88 GMT-4] [CPU: 426.9 MB]     [Avail: 301.04 GB] Input particles have pixel size 0.8300
> [2026-06-05 17:52:02.89 GMT-4] [CPU: 426.9 MB]     [Avail: 301.04 GB] Particles will be zeropadded/truncated to size 320 during
                                alignment
```

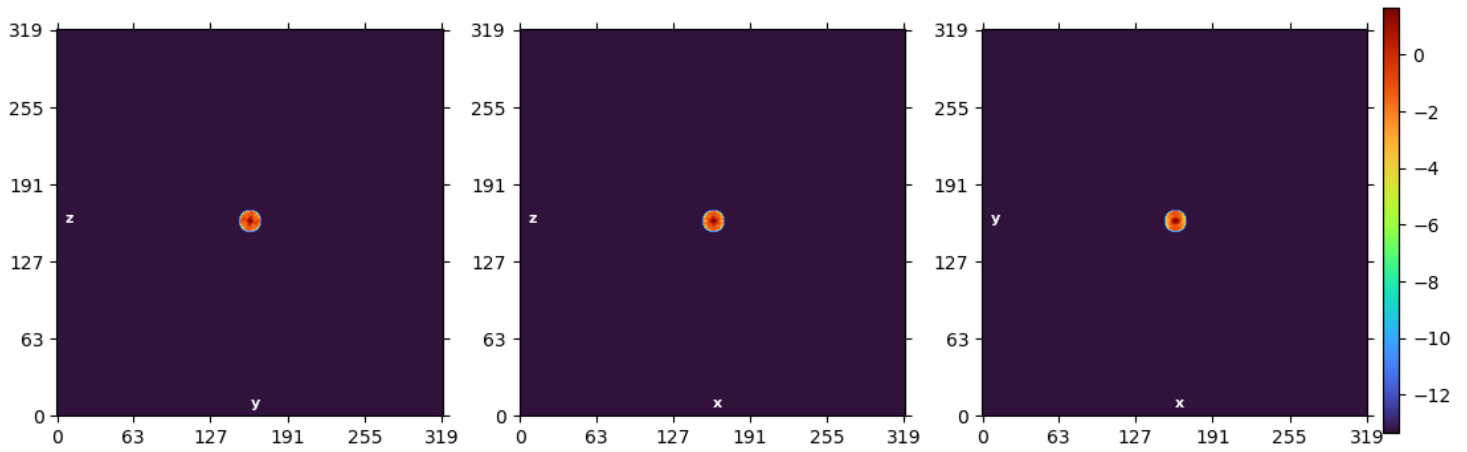
```

> [2026-06-05 17:52:02.89 GMT-4] [CPU: 426.9 MB] [Avail: 301.04 GB] Volume refinement will be done with effective box size 320
> [2026-06-05 17:52:02.89 GMT-4] [CPU: 426.9 MB] [Avail: 301.03 GB] Volume refinement will be done with pixel size 0.8300
> [2026-06-05 17:52:02.89 GMT-4] [CPU: 426.9 MB] [Avail: 301.03 GB] Particles will be zeropadded/truncated to size 320 during
backprojection
> [2026-06-05 17:52:02.90 GMT-4] [CPU: 426.9 MB] [Avail: 301.03 GB] Particles will be backprojected with box size 320
> [2026-06-05 17:52:02.90 GMT-4] [CPU: 426.9 MB] [Avail: 301.03 GB] Volume will be internally cropped and stored with box size
320
> [2026-06-05 17:52:02.90 GMT-4] [CPU: 427.0 MB] [Avail: 301.03 GB] Volume will be interpolated with box size 320 (zeropadding
factor 1.00)
> [2026-06-05 17:52:02.91 GMT-4] [CPU: 427.0 MB] [Avail: 301.03 GB] DC components of images will be ignored and volume will be
floated at each iteration.
> [2026-06-05 17:52:02.91 GMT-4] [CPU: 427.0 MB] [Avail: 301.03 GB] Spherical windowing of maps is enabled
> [2026-06-05 17:52:02.91 GMT-4] [CPU: 427.0 MB] [Avail: 301.03 GB] Refining with C1 symmetry enforced
> [2026-06-05 17:52:02.91 GMT-4] [CPU: 427.0 MB] [Avail: 301.03 GB] Refining with pose and shift marginalization during
backprojection enabled.
> [2026-06-05 17:52:02.92 GMT-4] [CPU: 427.0 MB] [Avail: 301.03 GB] Resetting input per-particle scale factors to 1.0
> [2026-06-05 17:52:02.92 GMT-4] [CPU: 427.0 MB] [Avail: 301.03 GB] Starting at initial resolution 30.000A (radwn 8.853).
> [2026-06-05 17:52:02.92 GMT-4] [CPU: 427.0 MB] [Avail: 301.04 GB] ===== Non-Uniform Refinement =====
> [2026-06-05 17:52:02.92 GMT-4] [CPU: 427.0 MB] [Avail: 301.04 GB] Non-Uniform Refinement is enabled.
> [2026-06-05 17:52:02.93 GMT-4] [CPU: 427.0 MB] [Avail: 301.04 GB] Using AWF of 3.00.
> [2026-06-05 17:52:02.93 GMT-4] [CPU: 427.0 MB] [Avail: 301.04 GB] Using butterworth filter with order 8.
> [2026-06-05 17:52:02.93 GMT-4] [CPU: 427.0 MB] [Avail: 301.04 GB] ===== Masking =====
> [2026-06-05 17:52:05.21 GMT-4] [CPU: 958.5 MB] [Avail: 303.44 GB] No mask input was connected, so dynamic masking will be
enabled.
> [2026-06-05 17:52:05.22 GMT-4] [CPU: 958.5 MB] [Avail: 303.44 GB] Dynamic mask threshold: 0.2000
> [2026-06-05 17:52:05.22 GMT-4] [CPU: 958.5 MB] [Avail: 303.43 GB] Dynamic mask near (A): 6.00
> [2026-06-05 17:52:05.22 GMT-4] [CPU: 958.5 MB] [Avail: 303.43 GB] Dynamic mask far (A): 14.00
> [2026-06-05 17:52:05.23 GMT-4] [CPU: 958.5 MB] [Avail: 303.42 GB] ===== Initial Model =====
> [2026-06-05 17:52:05.23 GMT-4] [CPU: 958.5 MB] [Avail: 303.42 GB] Resampling initial model to specified volume representation
size and pixel-size...
> [2026-06-05 17:52:08.26 GMT-4] [CPU: 1.35 GB] [Avail: 303.91 GB] Estimating scale of initial reference.
> [2026-06-05 17:52:13.95 GMT-4] [CPU: 77.4 MB] [Avail: 301.22 GB] WARNING: io_uring support disabled (not supported by kernel),
I/O performance may degrade
> [2026-06-05 17:52:14.90 GMT-4] [CPU: 1.56 GB] [Avail: 303.00 GB] Rescaling initial reference by a factor of 1.476
> [2026-06-05 17:52:16.59 GMT-4] [CPU: 1.56 GB] [Avail: 303.04 GB] Estimating scale of initial reference.
> [2026-06-05 17:52:19.64 GMT-4] [CPU: 1.51 GB] [Avail: 304.49 GB] Rescaling initial reference by a factor of 0.992
> [2026-06-05 17:52:21.24 GMT-4] [CPU: 1.51 GB] [Avail: 305.31 GB] Estimating scale of initial reference.
> [2026-06-05 17:52:24.28 GMT-4] [CPU: 1.51 GB] [Avail: 302.72 GB] Rescaling initial reference by a factor of 0.997
> [2026-06-05 17:52:27.57 GMT-4] Initial Real Space Slices

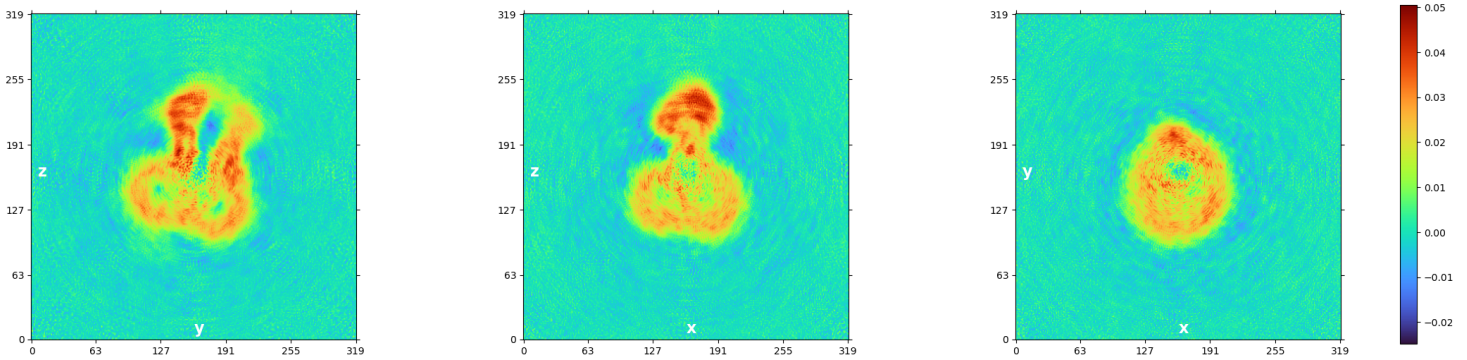
```



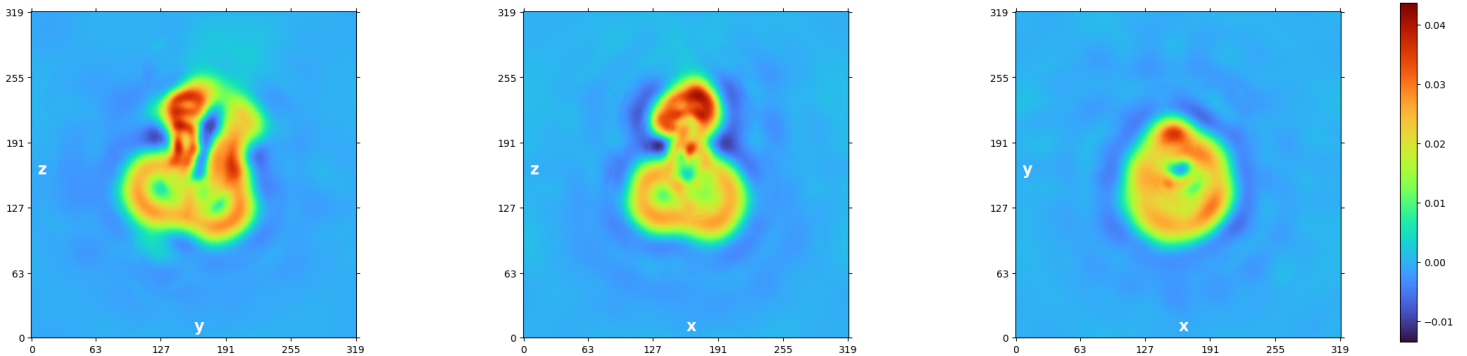
> [2026-06-05 17:52:29.11 GMT-4] Initial Fourier Space Slices



```
> [2026-06-05 17:52:29.11 GMT-4] [CPU: 1.56 GB] [Avail: 304.72 GB] ===== Starting Refinement Iterations =====
> [2026-06-05 17:52:29.11 GMT-4] [CPU: 1.56 GB] [Avail: 304.71 GB] ----- Start Iteration 0
> [2026-06-05 17:52:29.12 GMT-4] [CPU: 1.56 GB] [Avail: 304.71 GB] Using Max Alignment Radius 8.853 (30.000A)
> [2026-06-05 17:52:29.12 GMT-4] [CPU: 1.56 GB] [Avail: 304.70 GB] Auto batchsize: 5942 in each split
> [2026-06-05 17:52:32.85 GMT-4] [CPU: 2.07 GB] [Avail: 298.81 GB] -- THR 1 BATCH 500 NUM 1471 TOTAL 4.5813093 ELAPSED 32.233744
--
> [2026-06-05 17:53:07.22 GMT-4] [CPU: 3.22 GB] [Avail: 295.75 GB] Processed 11884.000 images in 34.369s.
> [2026-06-05 17:53:11.47 GMT-4] [CPU: 3.54 GB] [Avail: 295.71 GB] Computing FSCs...
> [2026-06-05 17:53:11.47 GMT-4] [CPU: 3.54 GB] [Avail: 295.71 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-05 17:53:11.47 GMT-4] [CPU: 3.54 GB] [Avail: 295.71 GB] Computing FFTs on GPU.
> [2026-06-05 17:53:14.14 GMT-4] [CPU: 3.79 GB] [Avail: 297.11 GB] Done in 2.666s
> [2026-06-05 17:53:14.14 GMT-4] [CPU: 3.79 GB] [Avail: 297.11 GB] Computing cFSCs...
> [2026-06-05 17:53:16.29 GMT-4] [CPU: 3.79 GB] [Avail: 297.86 GB] Done in 2.143s
> [2026-06-05 17:53:16.29 GMT-4] [CPU: 3.79 GB] [Avail: 297.92 GB] Using Filter Radius 27.924 (9.511A) | Previous: 8.853
(30.000A)
> [2026-06-05 17:53:22.44 GMT-4] [CPU: 4.87 GB] [Avail: 297.56 GB] Non-uniform regularization with compute option: GPU
> [2026-06-05 17:53:22.45 GMT-4] [CPU: 4.87 GB] [Avail: 297.56 GB] Running local cross validation for A ...
> [2026-06-05 17:53:53.10 GMT-4] [CPU: 7.54 GB] [Avail: 297.05 GB] Local cross validation A done in 30.653s
> [2026-06-05 17:53:54.42 GMT-4] FSC Filtered Side A
```

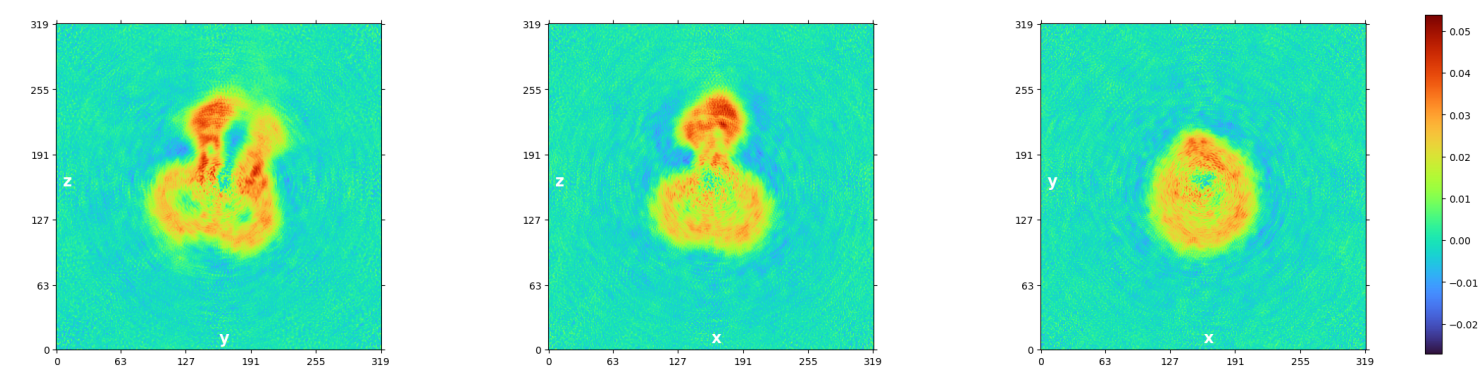


> [2026-06-05 17:53:55.51 GMT-4] CV Filtered Side A

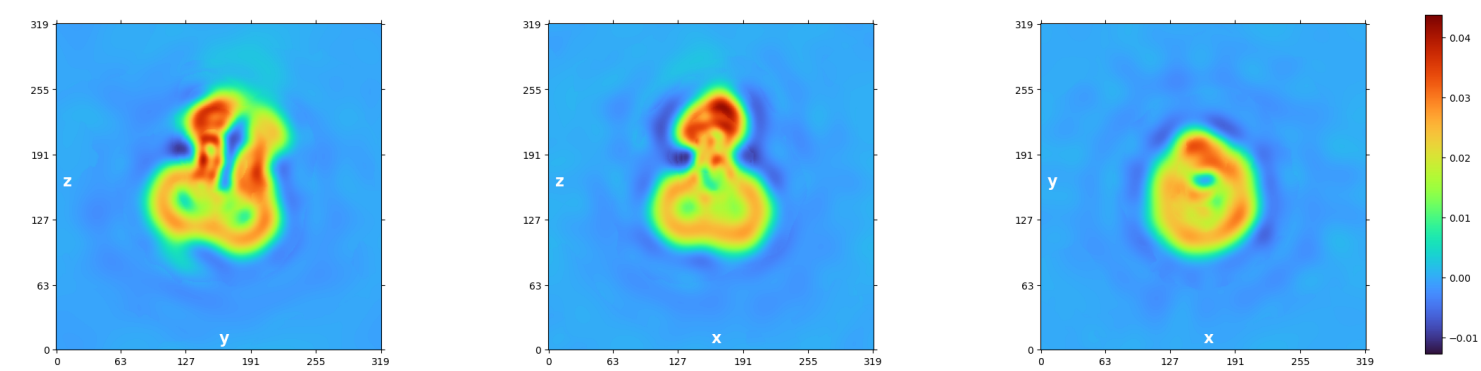


```
> [2026-06-05 17:53:55.51 GMT-4] [CPU: 7.55 GB] [Avail: 297.84 GB] Running local cross validation for B ...
> [2026-06-05 17:54:17.54 GMT-4] [CPU: 7.82 GB] [Avail: 294.44 GB] Local cross validation B done in 22.022s
```


> [2026-06-05 17:54:19.17 GMT-4] FSC Filtered Side B



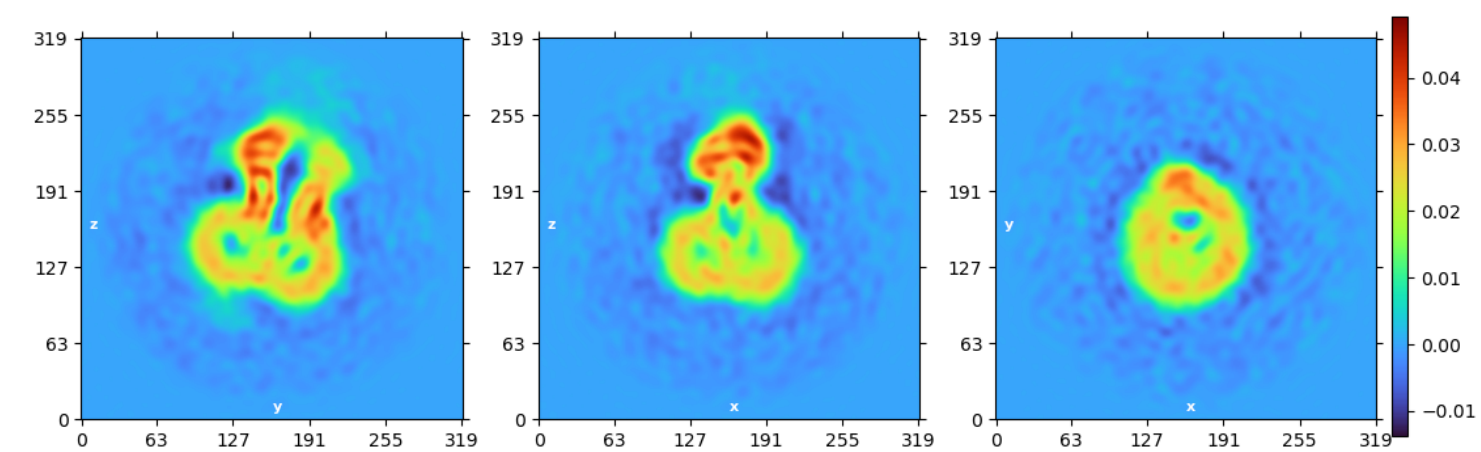
> [2026-06-05 17:54:20.51 GMT-4] CV Filtered Side B



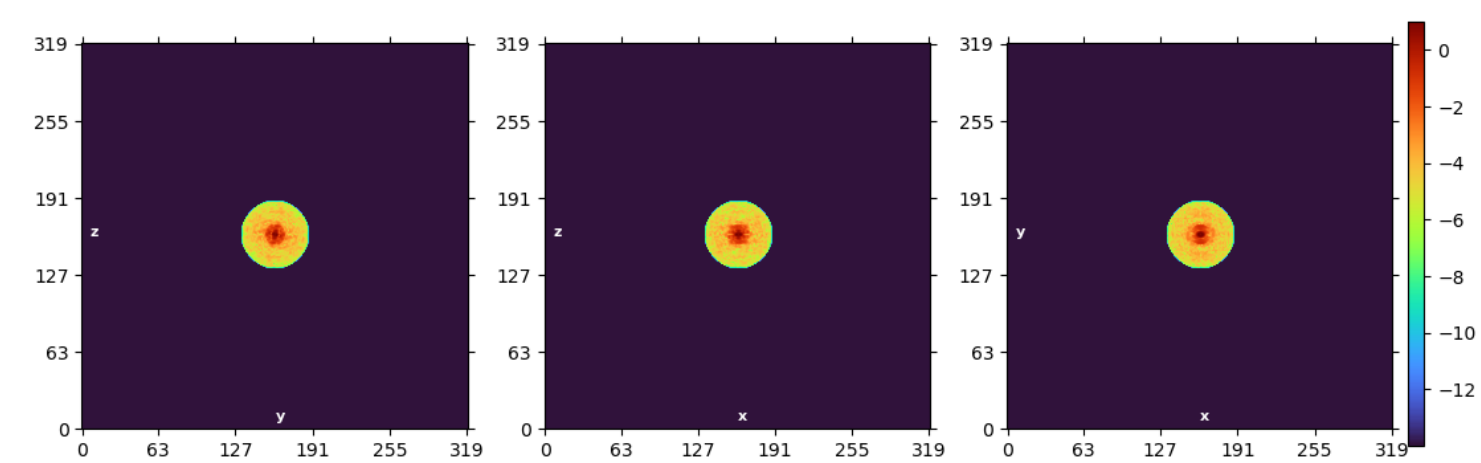
> [2026-06-05 17:54:28.84 GMT-4] [CPU: 8.15 GB] [Avail: 293.43 GB] Estimated Bfactor: -406.2

> [2026-06-05 17:54:28.85 GMT-4] [CPU: 8.15 GB] [Avail: 293.50 GB] Plotting..

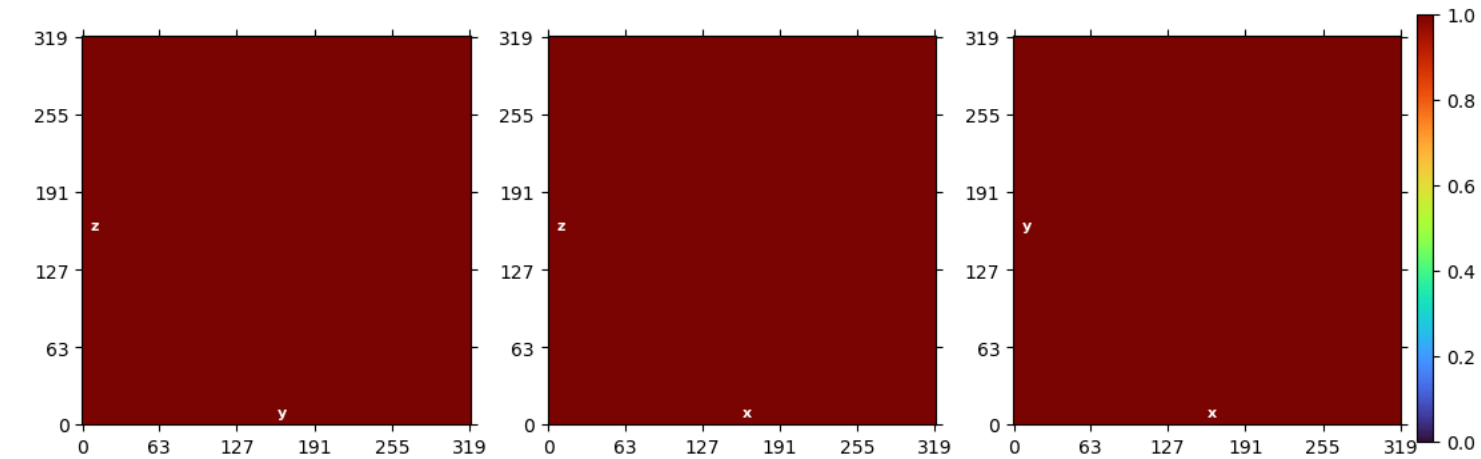
> [2026-06-05 17:54:35.10 GMT-4] Real Space Slices Iteration 000



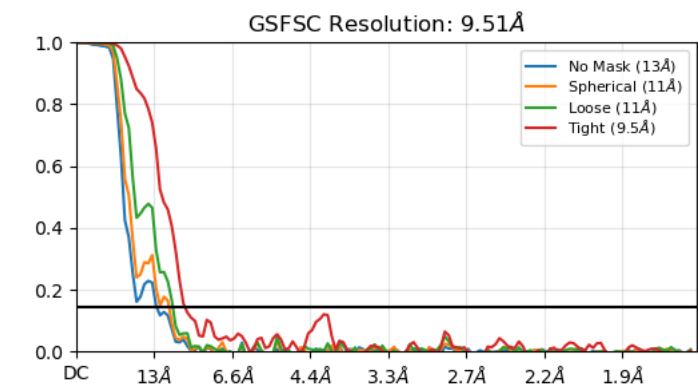
> [2026-06-05 17:54:36.41 GMT-4] Fourier Space Slices Iteration 000



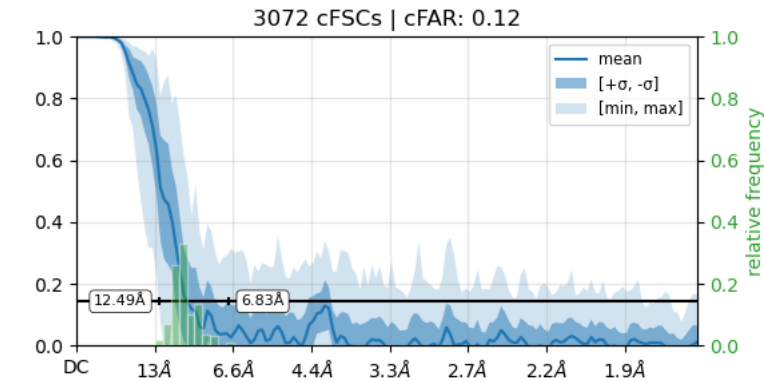
> [2026-06-05 17:54:37.52 GMT-4] Real Space Mask Slices Iteration 000



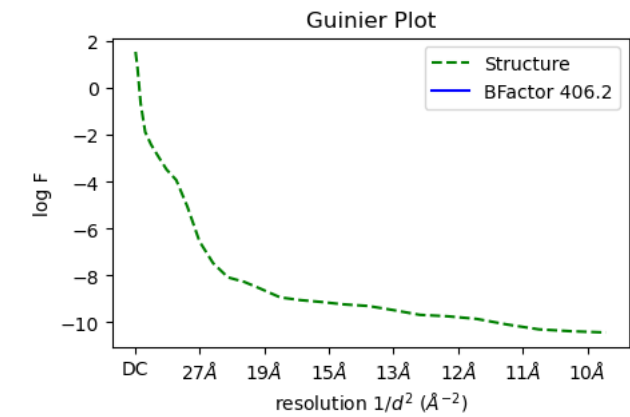
> [2026-06-05 17:54:38.24 GMT-4] FSC Iteration 000



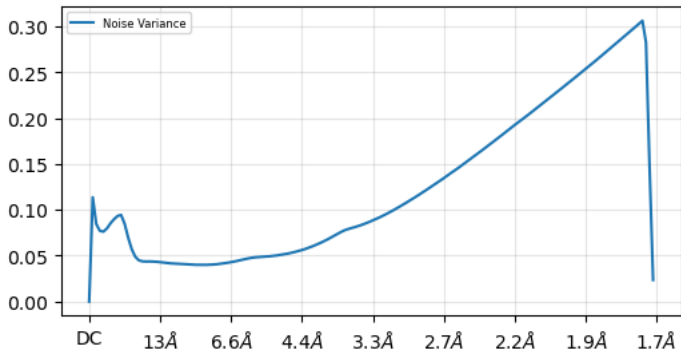
> [2026-06-05 17:54:39.09 GMT-4] cFSCs (Half-angle: 20°) Iteration 000, with tight mask



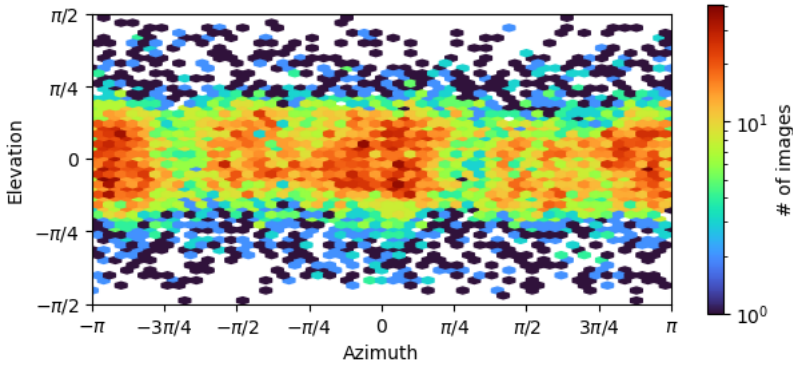
> [2026-06-05 17:54:44.54 GMT-4] Guinier Plot Iteration 000



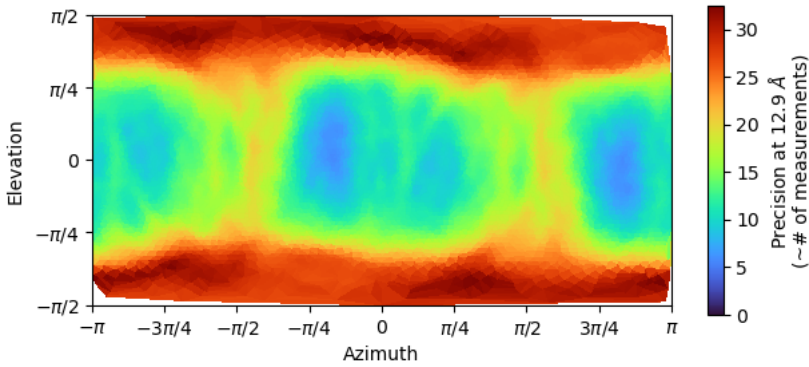
> [2026-06-05 17:54:44.87 GMT-4] Noise Model Iteration 000



> [2026-06-05 17:54:45.74 GMT-4] Viewing Direction Distribution Iteration 000

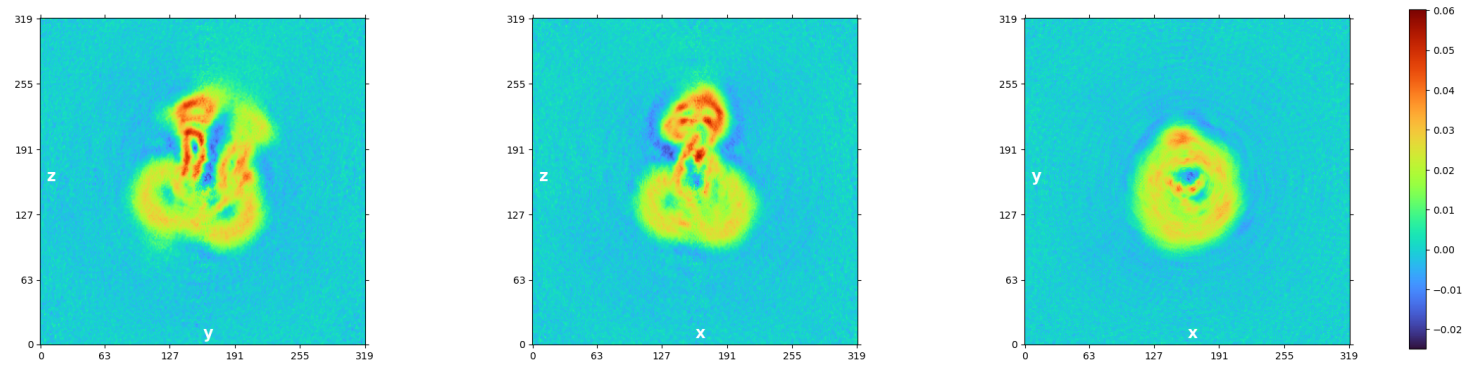


> [2026-06-05 17:54:47.65 GMT-4] Posterior Precision Directional Distribution Iteration 000

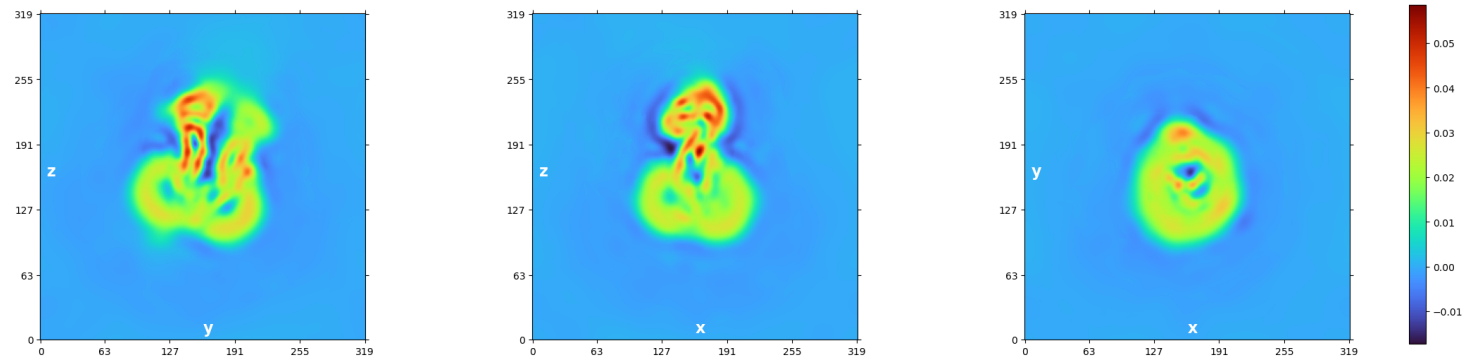


```
> [2026-06-05 17:54:47.65 GMT-4] [CPU: 8.57 GB] [Avail: 292.93 GB] Done in 18.802s.
> [2026-06-05 17:54:47.66 GMT-4] [CPU: 8.57 GB] [Avail: 292.93 GB] Outputting files..
> [2026-06-05 17:55:02.19 GMT-4] [CPU: 9.01 GB] [Avail: 294.11 GB] Done in 14.534s.
> [2026-06-05 17:55:02.20 GMT-4] [CPU: 9.01 GB] [Avail: 294.11 GB] Done iteration 0 in 153.081s. Total time so far 193.591s
> [2026-06-05 17:55:02.90 GMT-4] [CPU: 9.01 GB] [Avail: 295.76 GB] ----- Start Iteration 1
> [2026-06-05 17:55:02.90 GMT-4] [CPU: 9.01 GB] [Avail: 295.76 GB] Using Max Alignment Radius 27.924 (9.511A)
> [2026-06-05 17:55:02.91 GMT-4] [CPU: 9.01 GB] [Avail: 295.76 GB] Auto batchsize: 19117 in each split
> [2026-06-05 17:55:02.91 GMT-4] [CPU: 9.01 GB] [Avail: 295.76 GB] Using dynamic mask.
> [2026-06-05 17:55:31.80 GMT-4] [CPU: 9.51 GB] [Avail: 293.80 GB] -- THR 0 BATCH 500 NUM 5000 TOTAL 7.0057334 ELAPSED 72.120334
--
> [2026-06-05 17:56:46.58 GMT-4] [CPU: 10.39 GB] [Avail: 292.39 GB] Processed 38234.000 images in 74.776s.
> [2026-06-05 17:56:49.83 GMT-4] [CPU: 10.43 GB] [Avail: 293.62 GB] Computing FSCs...
> [2026-06-05 17:56:49.84 GMT-4] [CPU: 10.43 GB] [Avail: 293.62 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-05 17:56:49.84 GMT-4] [CPU: 10.43 GB] [Avail: 293.61 GB] Computing FFTs on GPU.
> [2026-06-05 17:56:51.76 GMT-4] [CPU: 10.43 GB] [Avail: 293.58 GB] Done in 1.919s
> [2026-06-05 17:56:51.76 GMT-4] [CPU: 10.43 GB] [Avail: 293.58 GB] Computing cFSCs...
> [2026-06-05 17:56:53.83 GMT-4] [CPU: 10.43 GB] [Avail: 293.45 GB] Done in 2.065s
> [2026-06-05 17:56:53.83 GMT-4] [CPU: 10.43 GB] [Avail: 293.45 GB] Using Filter Radius 43.643 (6.086A) | Previous: 27.924
(9.511A)
> [2026-06-05 17:57:00.67 GMT-4] [CPU: 9.67 GB] [Avail: 294.85 GB] Non-uniform regularization with compute option: GPU
> [2026-06-05 17:57:00.67 GMT-4] [CPU: 9.67 GB] [Avail: 294.84 GB] Running local cross validation for A ...
> [2026-06-05 17:57:23.69 GMT-4] [CPU: 9.89 GB] [Avail: 288.87 GB] Local cross validation A done in 23.008s
```


> [2026-06-05 17:57:25.33 GMT-4] FSC Filtered Side A



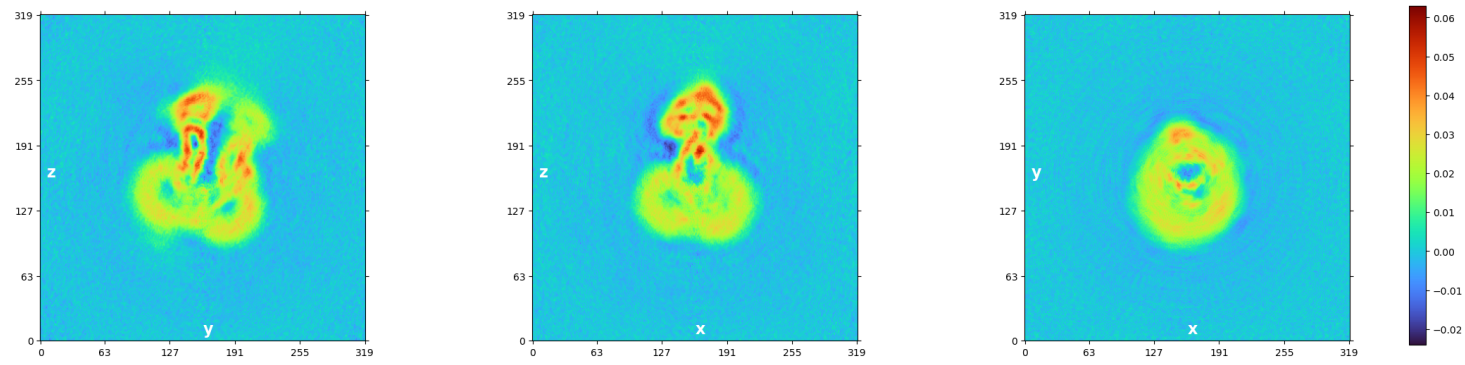
> [2026-06-05 17:57:26.42 GMT-4] CV Filtered Side A



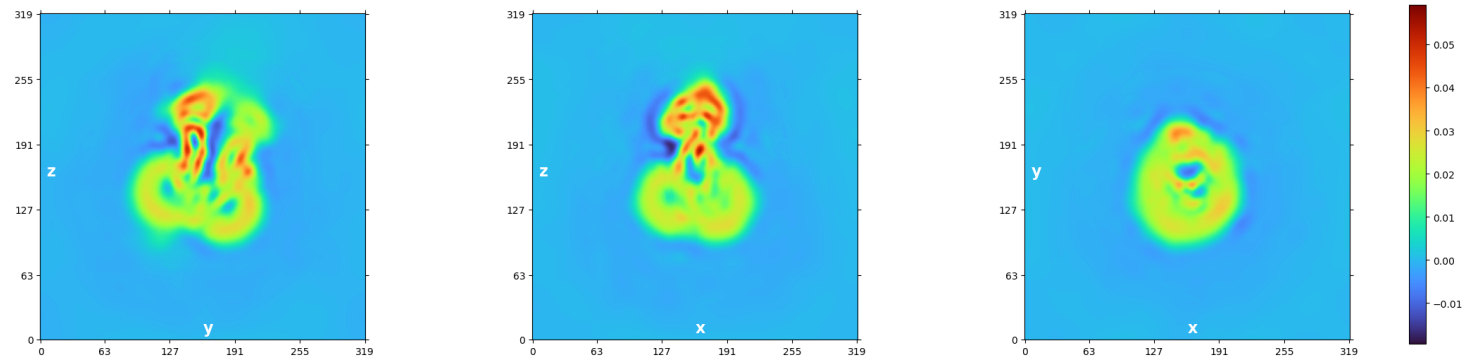
> [2026-06-05 17:57:26.42 GMT-4] [CPU: 9.89 GB] [Avail: 288.22 GB] Running local cross validation for B ...

> [2026-06-05 17:57:48.93 GMT-4] [CPU: 10.14 GB] [Avail: 290.70 GB] Local cross validation B done in 22.500s

> [2026-06-05 17:57:50.49 GMT-4] FSC Filtered Side B



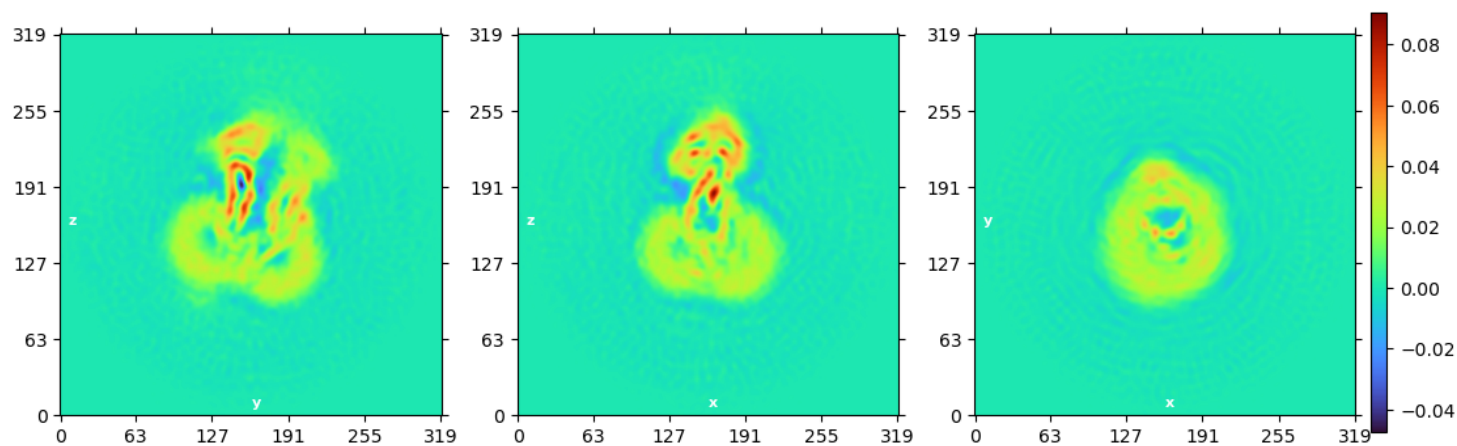
> [2026-06-05 17:57:51.68 GMT-4] CV Filtered Side B



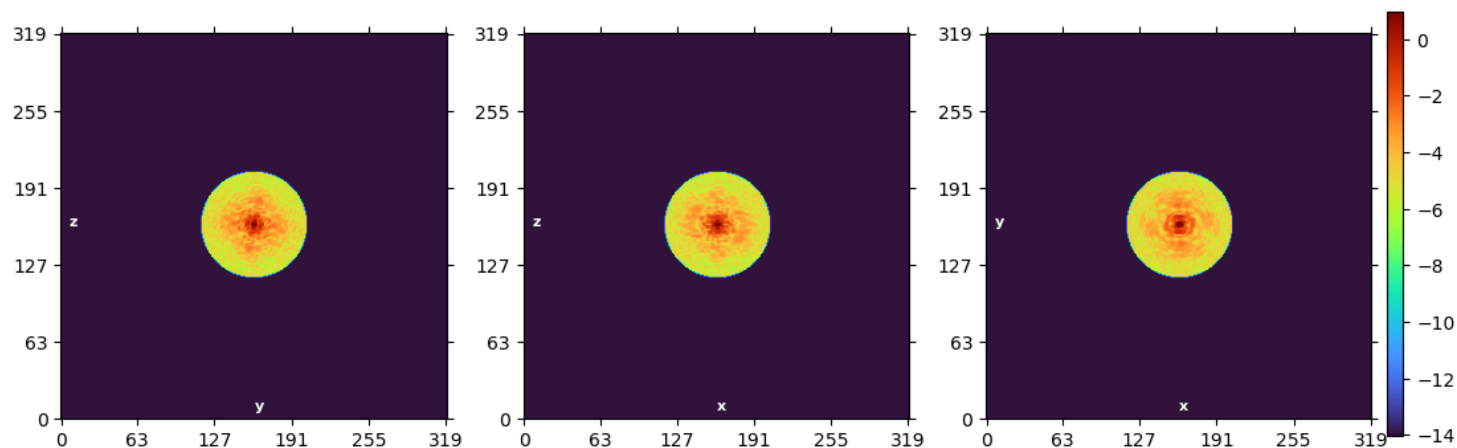
> [2026-06-05 17:58:03.31 GMT-4] [CPU: 10.36 GB] [Avail: 304.69 GB] Estimated Bfactor: -361.1

> [2026-06-05 17:58:03.32 GMT-4] [CPU: 10.36 GB] [Avail: 304.69 GB] Plotting..

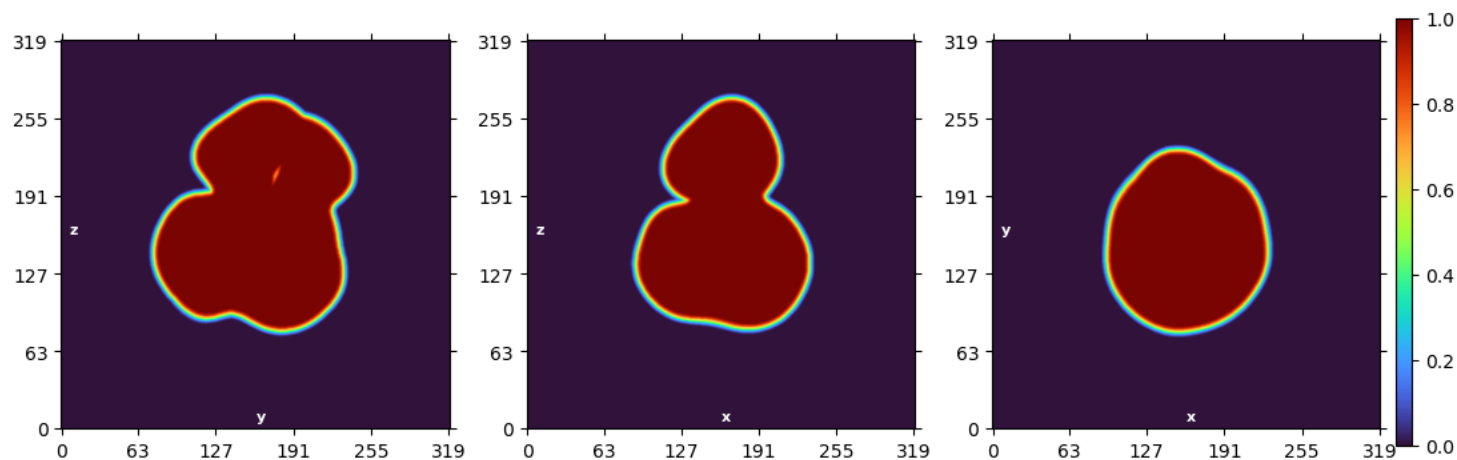
> [2026-06-05 17:58:10.38 GMT-4] Real Space Slices Iteration 001



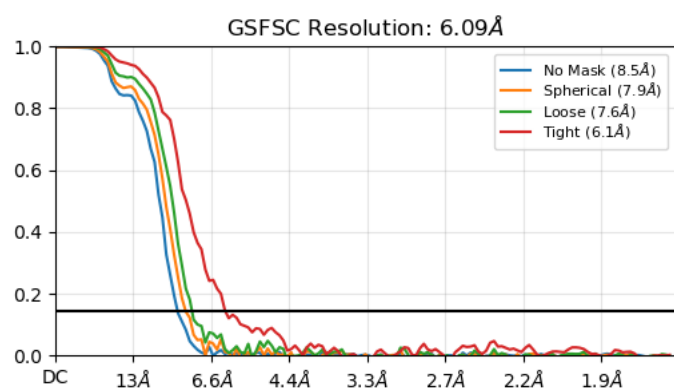
> [2026-06-05 17:58:11.72 GMT-4] Fourier Space Slices Iteration 001



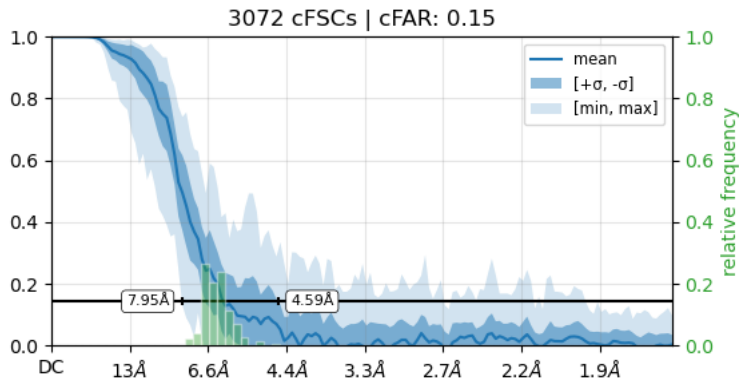
> [2026-06-05 17:58:13.74 GMT-4] Real Space Mask Slices Iteration 001



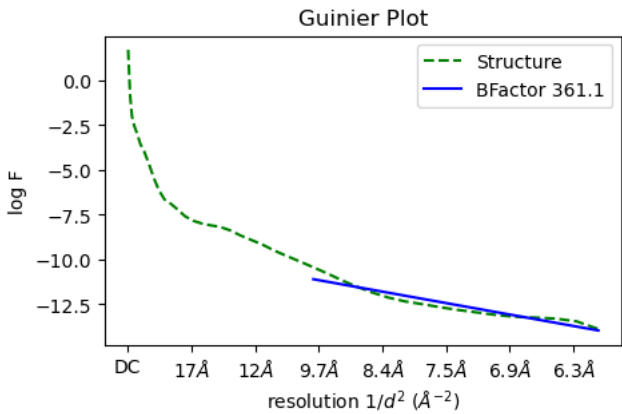
> [2026-06-05 17:58:14.30 GMT-4] FSC Iteration 001



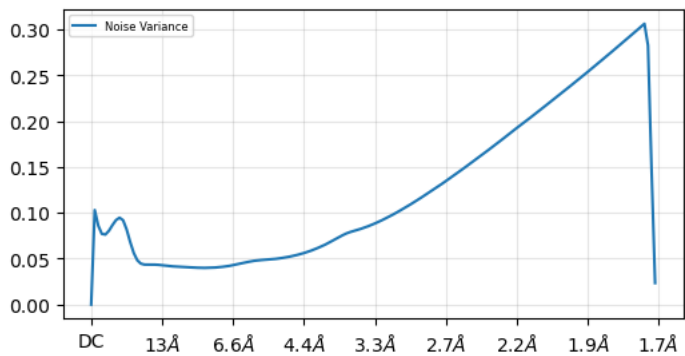
> [2026-06-05 17:58:15.10 GMT-4] cFSCs (Half-angle: 20°) Iteration 001, with tight mask



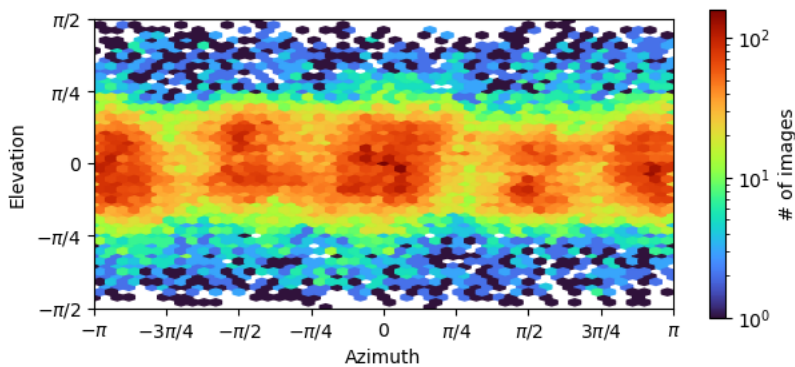
> [2026-06-05 17:58:21.16 GMT-4] Guinier Plot Iteration 001



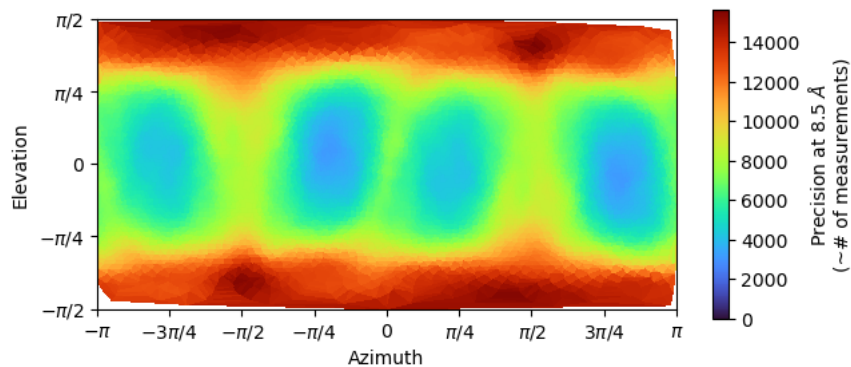
> [2026-06-05 17:58:21.58 GMT-4] Noise Model Iteration 001



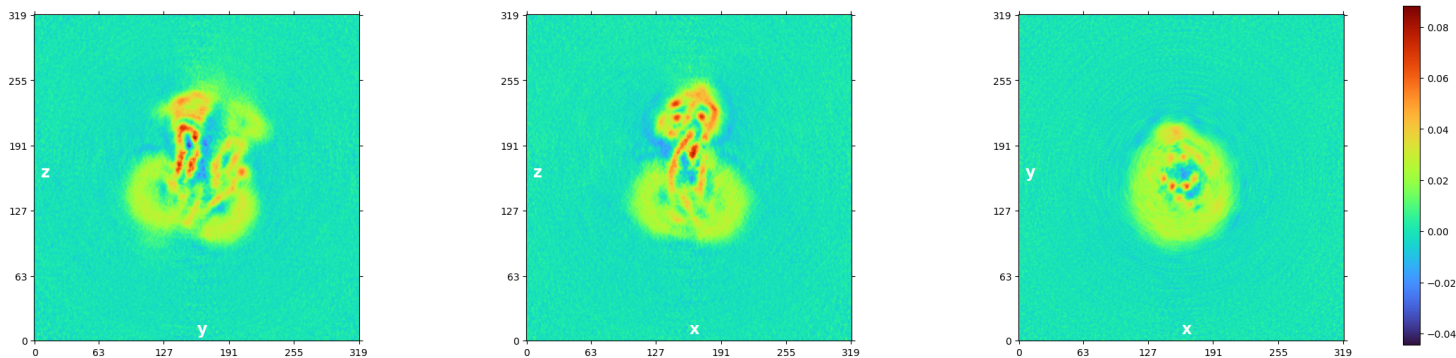
> [2026-06-05 17:58:22.85 GMT-4] Viewing Direction Distribution Iteration 001



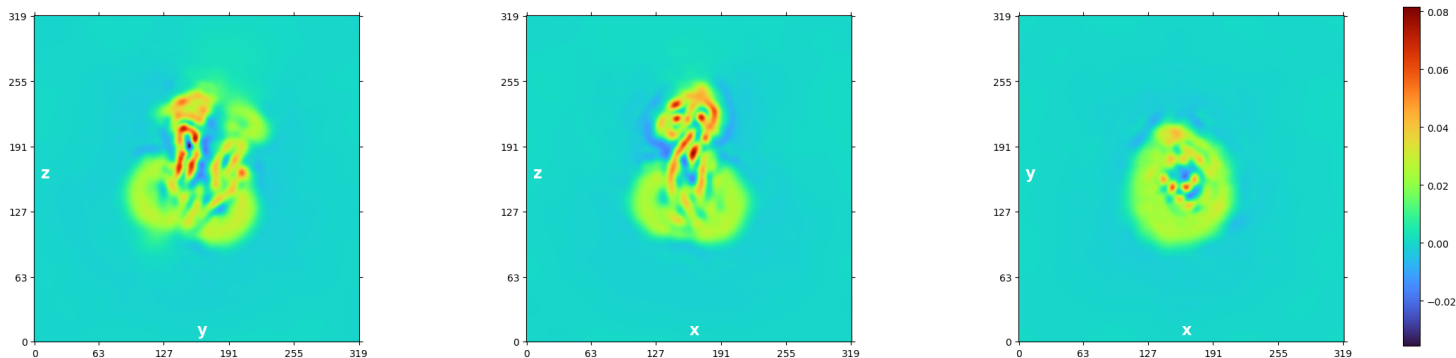
> [2026-06-05 17:58:24.30 GMT-4] Posterior Precision Directional Distribution Iteration 001



```
> [2026-06-05 17:58:24.30 GMT-4] [CPU: 10.39 GB] [Avail: 304.83 GB] Done in 20.977s.
> [2026-06-05 17:58:24.31 GMT-4] [CPU: 10.39 GB] [Avail: 304.86 GB] Outputting files..
> [2026-06-05 17:58:41.40 GMT-4] [CPU: 10.33 GB] [Avail: 300.77 GB] Done in 17.094s.
> [2026-06-05 17:58:41.41 GMT-4] [CPU: 10.33 GB] [Avail: 300.77 GB] Done iteration 1 in 218.503s. Total time so far 412.801s
> [2026-06-05 17:58:42.25 GMT-4] [CPU: 10.33 GB] [Avail: 300.56 GB] ----- Start Iteration 2
> [2026-06-05 17:58:42.25 GMT-4] [CPU: 10.33 GB] [Avail: 300.55 GB] Using Max Alignment Radius 43.643 (6.086A)
> [2026-06-05 17:58:42.25 GMT-4] [CPU: 10.33 GB] [Avail: 300.54 GB] Auto batchsize: 29975 in each split
> [2026-06-05 17:58:42.26 GMT-4] [CPU: 10.33 GB] [Avail: 300.54 GB] Using dynamic mask.
> [2026-06-05 17:59:12.74 GMT-4] [CPU: 10.33 GB] [Avail: 298.60 GB] -- THR 0 BATCH 500 NUM 7488 TOTAL 10.464320 ELAPSED 106.99147
--
> [2026-06-05 18:01:02.24 GMT-4] [CPU: 11.19 GB] [Avail: 291.78 GB] Processed 59950.000 images in 109.502s.
> [2026-06-05 18:01:05.53 GMT-4] [CPU: 11.19 GB] [Avail: 296.18 GB] Computing FSCs...
> [2026-06-05 18:01:05.54 GMT-4] [CPU: 11.19 GB] [Avail: 296.18 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-05 18:01:05.54 GMT-4] [CPU: 11.19 GB] [Avail: 296.17 GB] Computing FFTs on GPU.
> [2026-06-05 18:01:07.83 GMT-4] [CPU: 11.19 GB] [Avail: 295.15 GB] Done in 2.296s
> [2026-06-05 18:01:07.84 GMT-4] [CPU: 11.19 GB] [Avail: 295.15 GB] Computing cFSCs...
> [2026-06-05 18:01:10.24 GMT-4] [CPU: 11.20 GB] [Avail: 295.37 GB] Done in 2.395s
> [2026-06-05 18:01:10.24 GMT-4] [CPU: 11.20 GB] [Avail: 295.36 GB] Using Filter Radius 61.097 (4.347A) | Previous: 43.643
(6.086A)
> [2026-06-05 18:01:17.11 GMT-4] [CPU: 10.44 GB] [Avail: 295.86 GB] Non-uniform regularization with compute option: GPU
> [2026-06-05 18:01:17.11 GMT-4] [CPU: 10.44 GB] [Avail: 295.89 GB] Running local cross validation for A ...
> [2026-06-05 18:01:40.22 GMT-4] [CPU: 10.69 GB] [Avail: 291.04 GB] Local cross validation A done in 23.108s
> [2026-06-05 18:01:41.96 GMT-4] FSC Filtered Side A
```



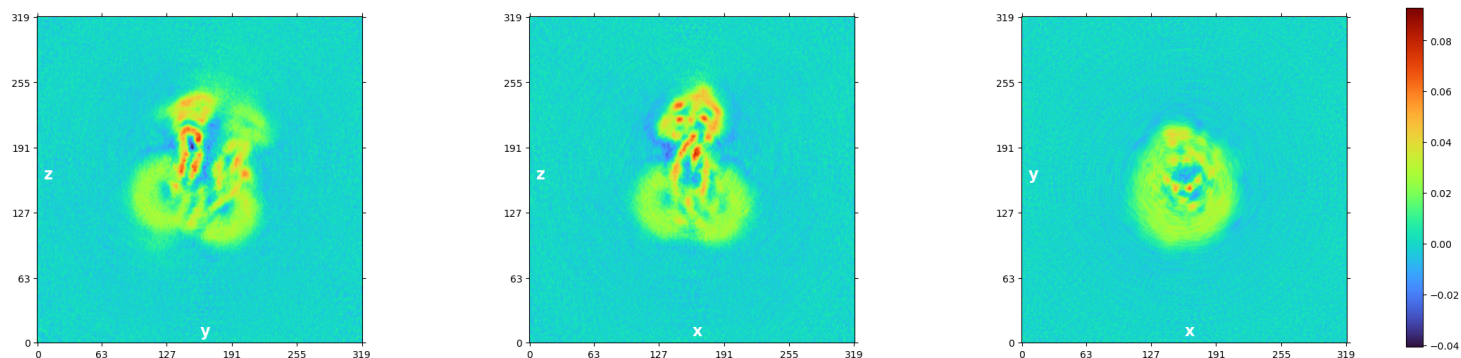
> [2026-06-05 18:01:43.24 GMT-4] CV Filtered Side A



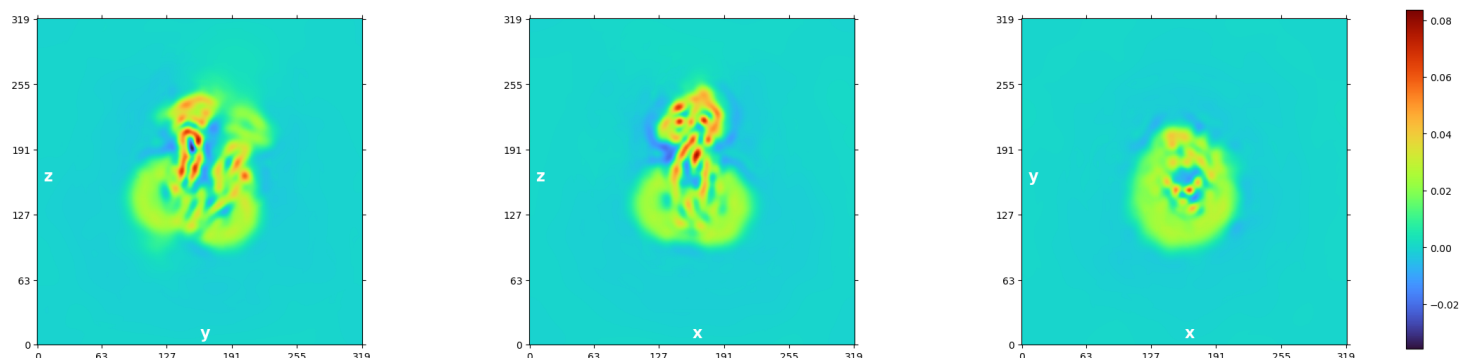
> [2026-06-05 18:01:43.24 GMT-4] [CPU: 10.69 GB] [Avail: 291.06 GB] Running local cross validation for B ...

> [2026-06-05 18:02:06.93 GMT-4] [CPU: 10.94 GB] [Avail: 289.51 GB] Local cross validation B done in 23.687s

> [2026-06-05 18:02:08.57 GMT-4] FSC Filtered Side B



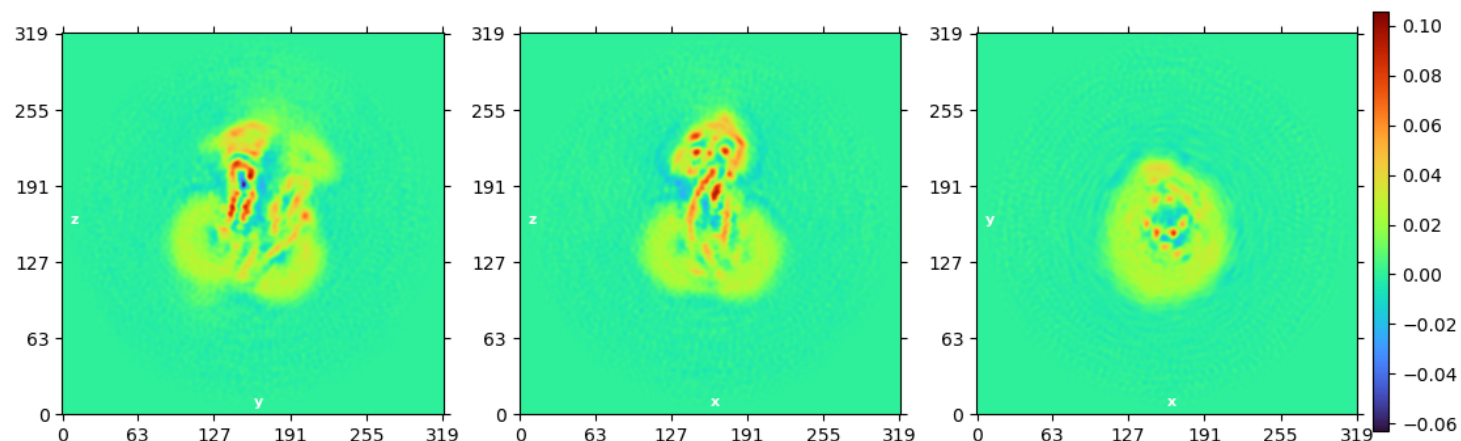
> [2026-06-05 18:02:09.79 GMT-4] CV Filtered Side B



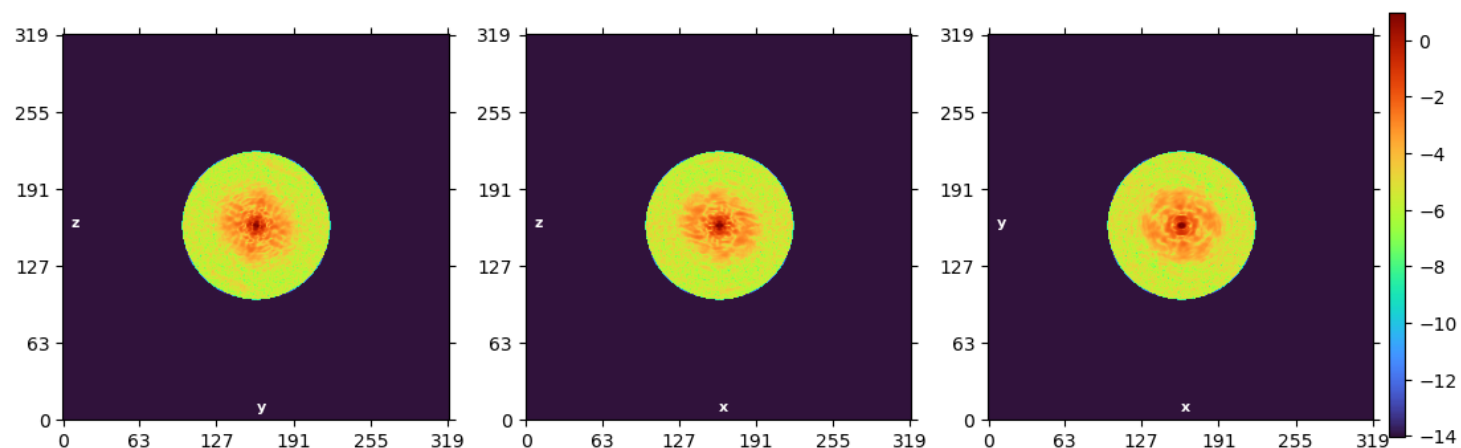
> [2026-06-05 18:02:20.12 GMT-4] [CPU: 10.93 GB] [Avail: 290.87 GB] Estimated Bfactor: -165.5

> [2026-06-05 18:02:20.13 GMT-4] [CPU: 10.93 GB] [Avail: 290.86 GB] Plotting..

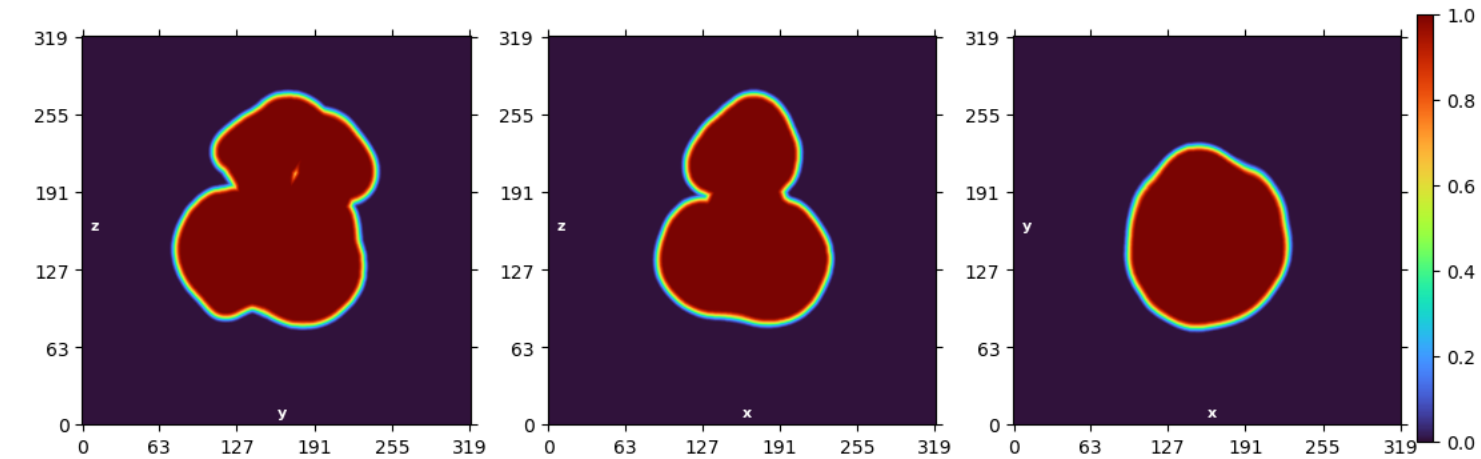
> [2026-06-05 18:02:26.41 GMT-4] Real Space Slices Iteration 002



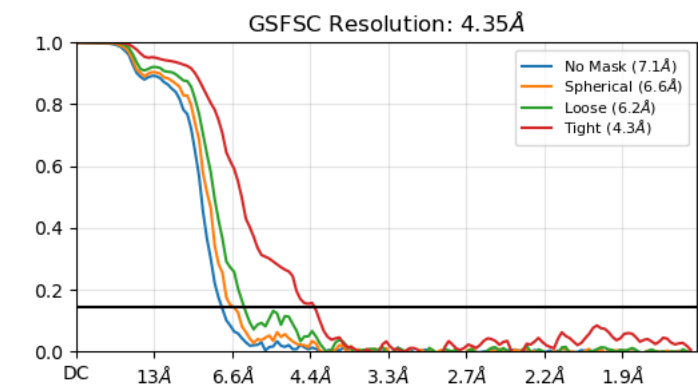
> [2026-06-05 18:02:27.90 GMT-4] Fourier Space Slices Iteration 002



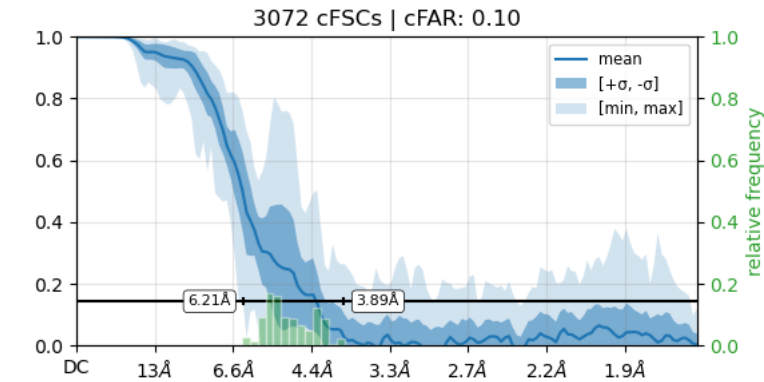
> [2026-06-05 18:02:29.73 GMT-4] Real Space Mask Slices Iteration 002



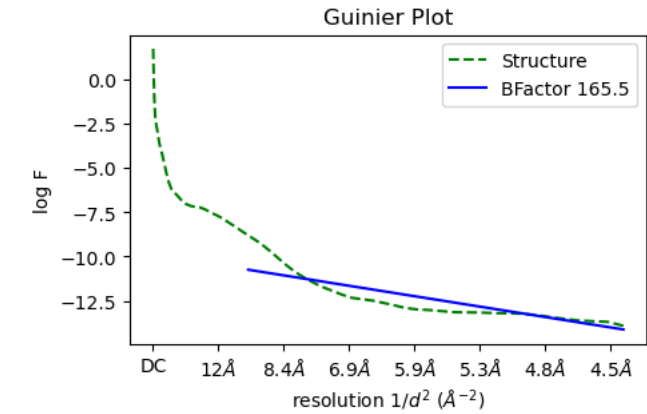
> [2026-06-05 18:02:30.25 GMT-4] FSC Iteration 002



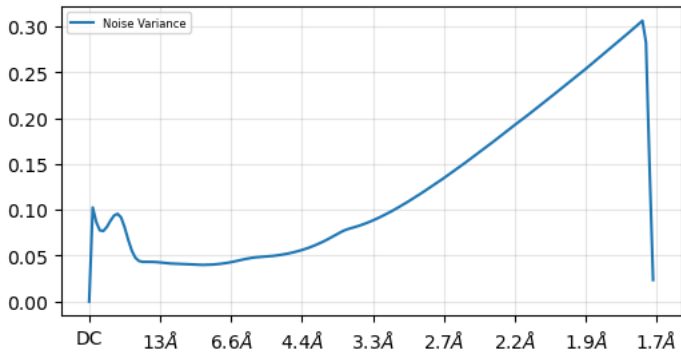
> [2026-06-05 18:02:31.02 GMT-4] cFSCs (Half-angle: 20°) Iteration 002, with tight mask



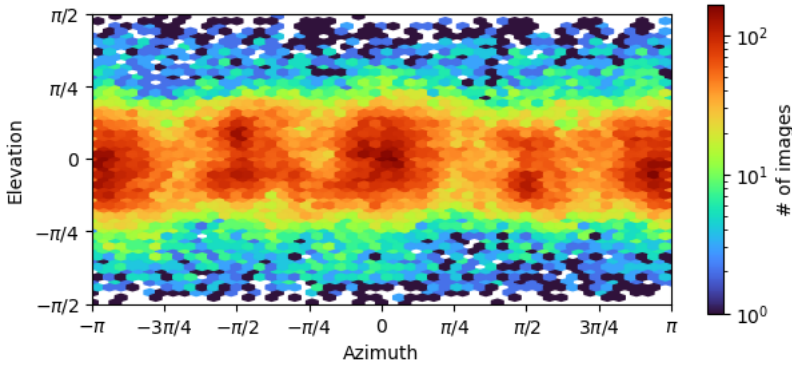
> [2026-06-05 18:02:36.85 GMT-4] Guinier Plot Iteration 002



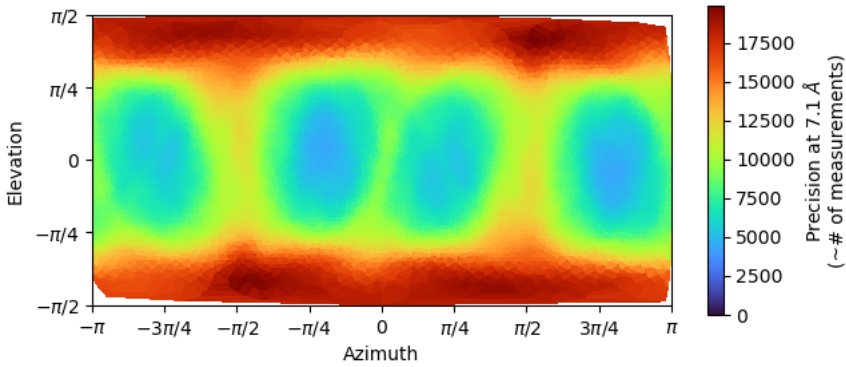
> [2026-06-05 18:02:37.24 GMT-4] Noise Model Iteration 002



> [2026-06-05 18:02:38.50 GMT-4] Viewing Direction Distribution Iteration 002

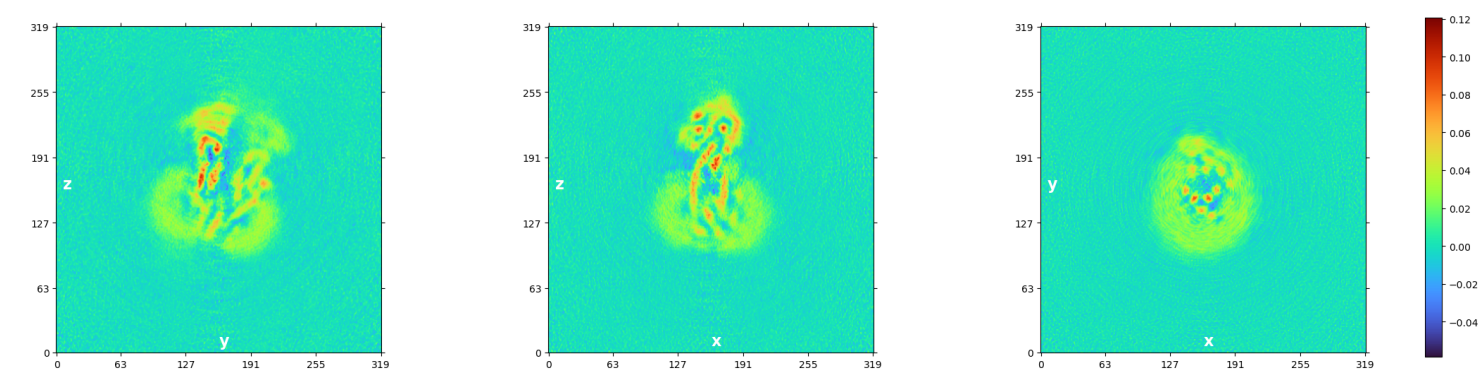


> [2026-06-05 18:02:40.20 GMT-4] Posterior Precision Directional Distribution Iteration 002

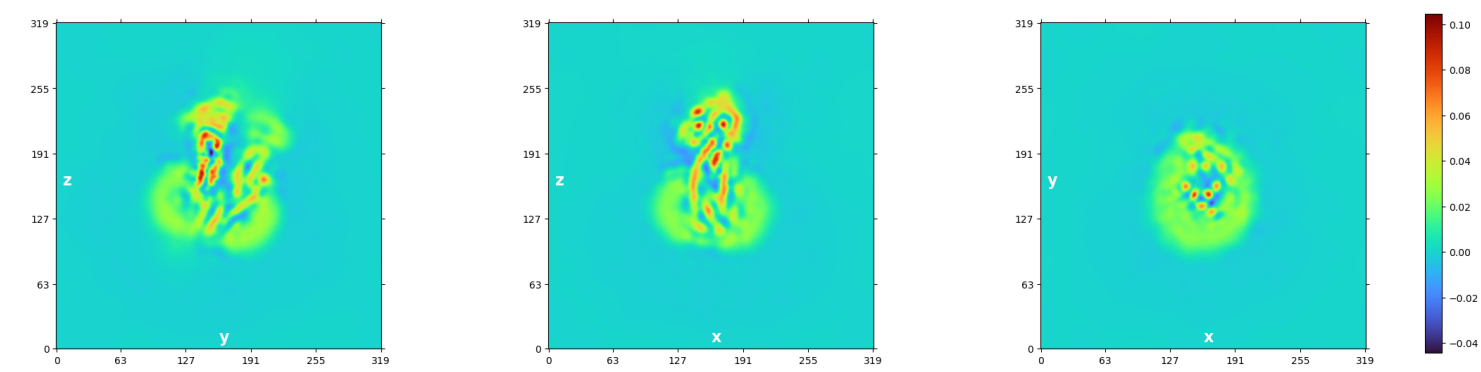


```
> [2026-06-05 18:02:40.20 GMT-4] [CPU: 10.95 GB] [Avail: 292.57 GB] Done in 20.069s.
> [2026-06-05 18:02:40.21 GMT-4] [CPU: 10.95 GB] [Avail: 292.57 GB] Outputting files..
> [2026-06-05 18:02:58.08 GMT-4] [CPU: 10.85 GB] [Avail: 296.21 GB] Done in 17.866s.
> [2026-06-05 18:02:58.08 GMT-4] [CPU: 10.85 GB] [Avail: 296.21 GB] Done iteration 2 in 255.831s. Total time so far 669.474s
> [2026-06-05 18:02:58.76 GMT-4] [CPU: 10.85 GB] [Avail: 296.19 GB] ----- Start Iteration 3
> [2026-06-05 18:02:58.76 GMT-4] [CPU: 10.85 GB] [Avail: 296.19 GB] Using Max Alignment Radius 61.097 (4.347A)
> [2026-06-05 18:02:58.76 GMT-4] [CPU: 10.85 GB] [Avail: 296.19 GB] Auto batchsize: 32724 in each split
> [2026-06-05 18:02:58.77 GMT-4] [CPU: 10.85 GB] [Avail: 296.19 GB] Using dynamic mask.
> [2026-06-05 18:03:27.21 GMT-4] [CPU: 10.85 GB] [Avail: 289.61 GB] -- THR 0 BATCH 500 NUM 8362 TOTAL 9.8387970 ELAPSED 115.78617
--
> [2026-06-05 18:05:25.15 GMT-4] [CPU: 11.61 GB] [Avail: 304.40 GB] Processed 65008.000 images in 117.947s.
> [2026-06-05 18:05:28.63 GMT-4] [CPU: 11.64 GB] [Avail: 305.21 GB] Computing FSCs...
> [2026-06-05 18:05:28.64 GMT-4] [CPU: 11.64 GB] [Avail: 305.19 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-05 18:05:28.64 GMT-4] [CPU: 11.64 GB] [Avail: 305.18 GB] Computing FFTs on GPU.
> [2026-06-05 18:05:30.73 GMT-4] [CPU: 11.64 GB] [Avail: 303.95 GB] Done in 2.089s
> [2026-06-05 18:05:30.73 GMT-4] [CPU: 11.64 GB] [Avail: 303.93 GB] Computing cFSCs...
> [2026-06-05 18:05:32.84 GMT-4] [CPU: 11.65 GB] [Avail: 304.66 GB] Done in 2.106s
> [2026-06-05 18:05:32.85 GMT-4] [CPU: 11.65 GB] [Avail: 304.71 GB] Using Filter Radius 64.746 (4.102A) | Previous: 61.097
(4.347A)
> [2026-06-05 18:05:39.57 GMT-4] [CPU: 10.89 GB] [Avail: 305.39 GB] Non-uniform regularization with compute option: GPU
> [2026-06-05 18:05:39.57 GMT-4] [CPU: 10.89 GB] [Avail: 305.38 GB] Running local cross validation for A ...
> [2026-06-05 18:06:02.72 GMT-4] [CPU: 11.11 GB] [Avail: 305.05 GB] Local cross validation A done in 23.140s
```


> [2026-06-05 18:06:04.24 GMT-4] FSC Filtered Side A



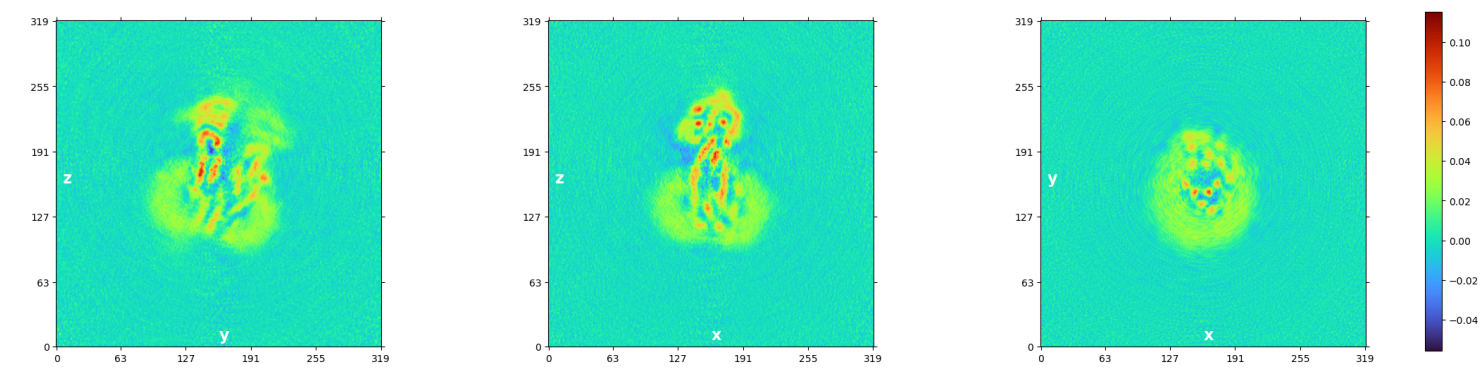
> [2026-06-05 18:06:05.52 GMT-4] CV Filtered Side A



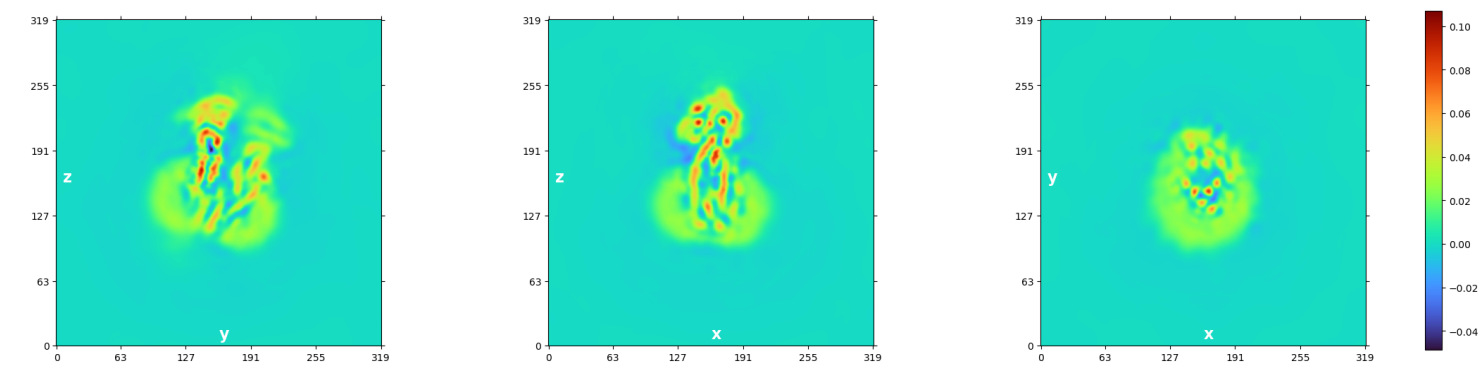
> [2026-06-05 18:06:05.52 GMT-4] [CPU: 11.11 GB] [Avail: 305.18 GB] Running local cross validation for B ...

> [2026-06-05 18:06:28.92 GMT-4] [CPU: 11.36 GB] [Avail: 316.01 GB] Local cross validation B done in 23.393s

> [2026-06-05 18:06:30.50 GMT-4] FSC Filtered Side B



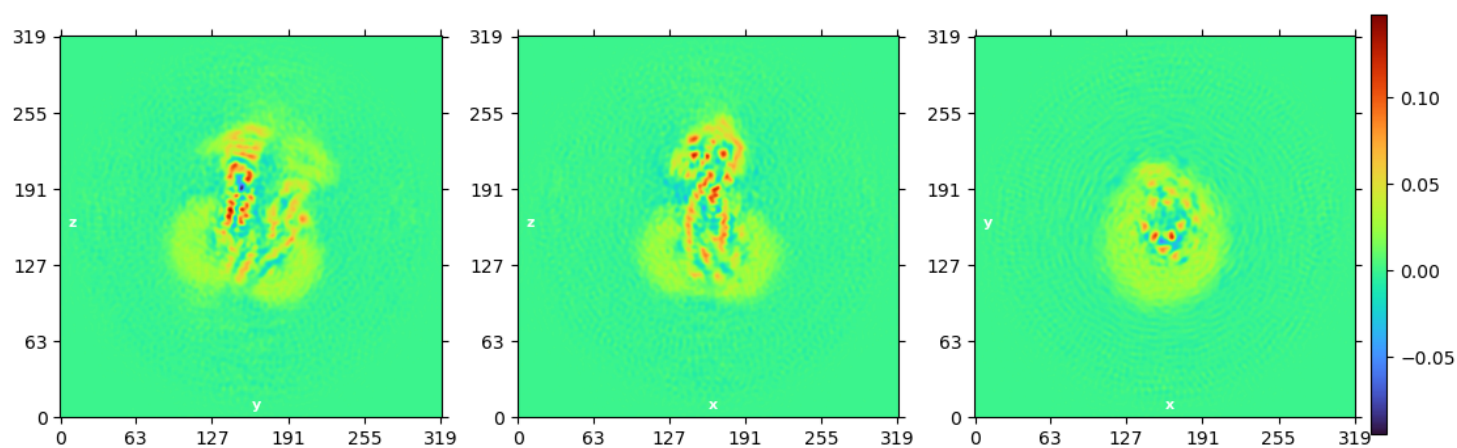
> [2026-06-05 18:06:31.60 GMT-4] CV Filtered Side B



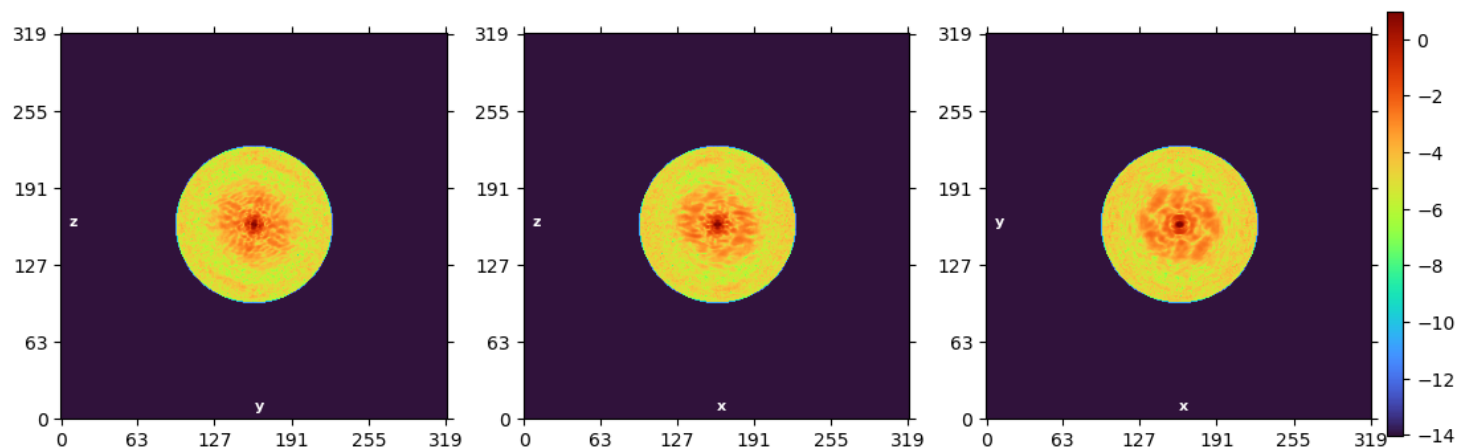
> [2026-06-05 18:06:41.85 GMT-4] [CPU: 11.38 GB] [Avail: 314.52 GB] Estimated Bfactor: -149.8

> [2026-06-05 18:06:41.85 GMT-4] [CPU: 11.38 GB] [Avail: 314.50 GB] Plotting..

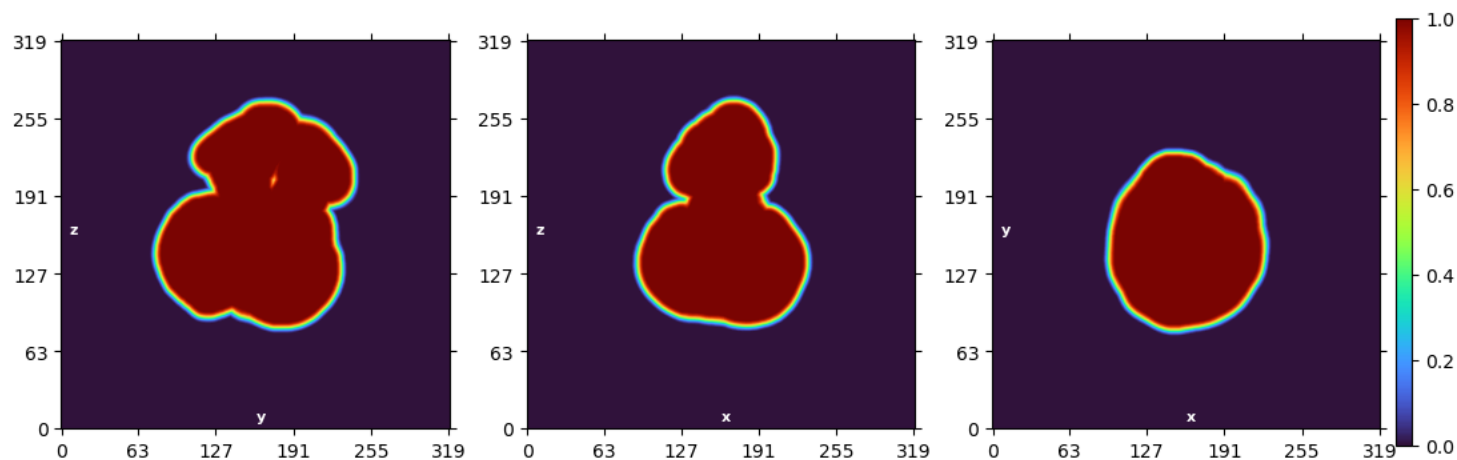
> [2026-06-05 18:06:48.19 GMT-4] Real Space Slices Iteration 003



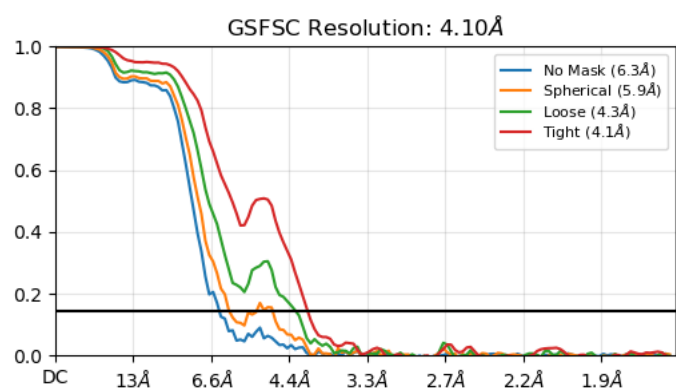
> [2026-06-05 18:06:49.48 GMT-4] Fourier Space Slices Iteration 003



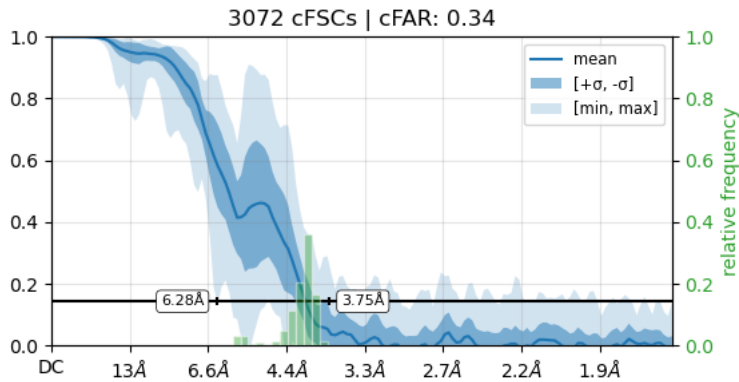
> [2026-06-05 18:06:51.46 GMT-4] Real Space Mask Slices Iteration 003



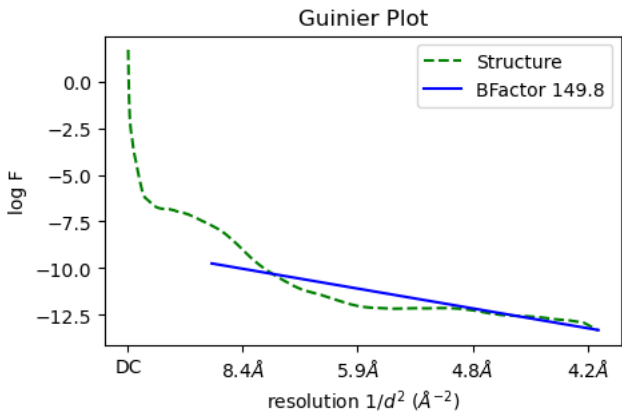
> [2026-06-05 18:06:52.10 GMT-4] FSC Iteration 003



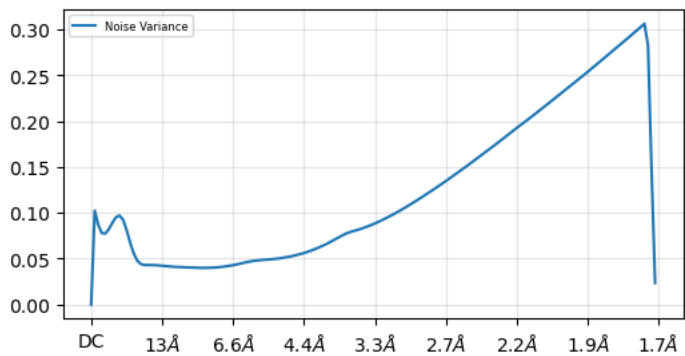
> [2026-06-05 18:06:52.96 GMT-4] cFSCs (Half-angle: 20°) Iteration 003, with tight mask



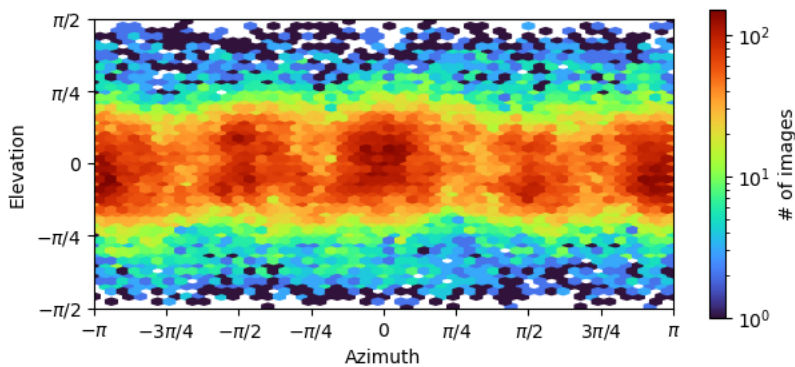
> [2026-06-05 18:06:59.04 GMT-4] Guinier Plot Iteration 003



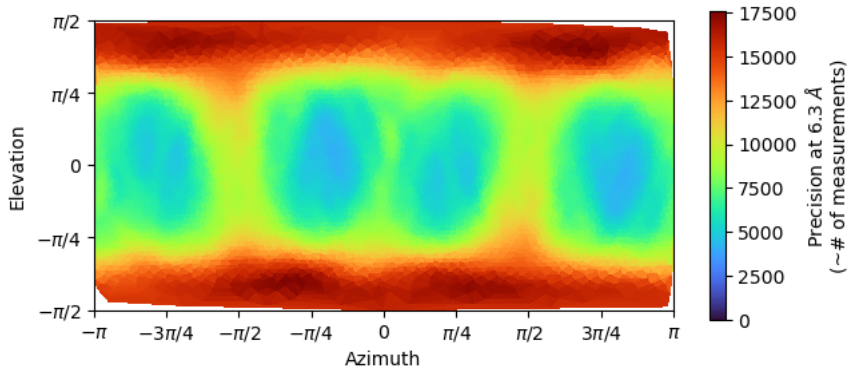
> [2026-06-05 18:06:59.41 GMT-4] Noise Model Iteration 003



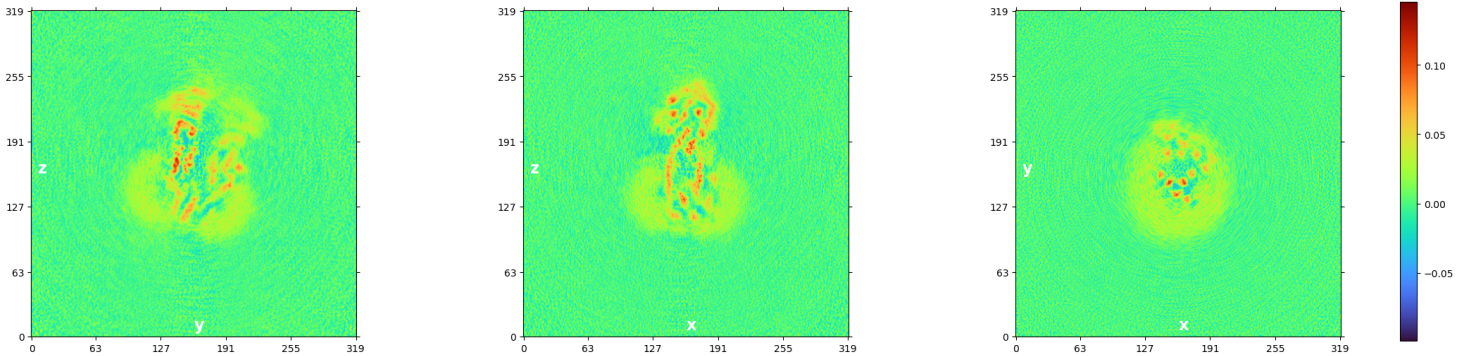
> [2026-06-05 18:07:00.63 GMT-4] Viewing Direction Distribution Iteration 003



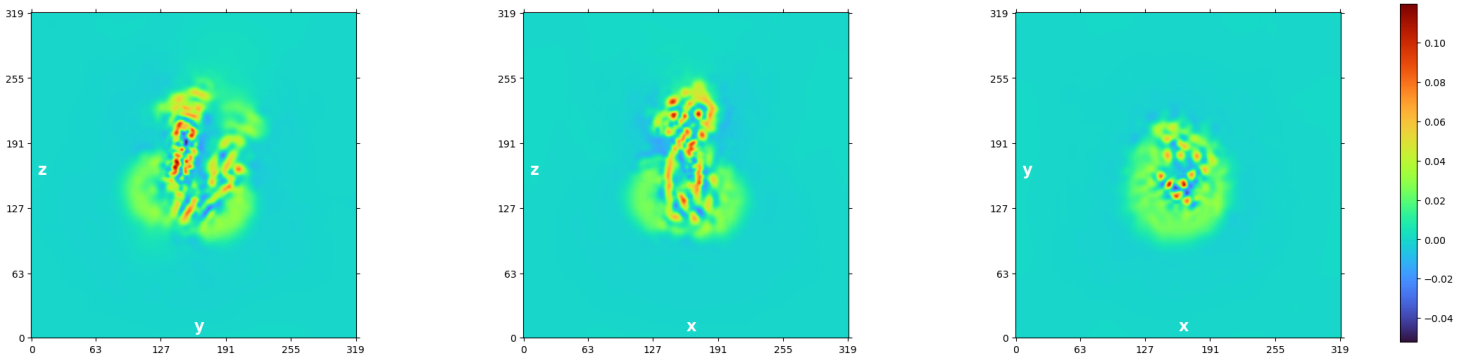
> [2026-06-05 18:07:02.27 GMT-4] Posterior Precision Directional Distribution Iteration 003



```
> [2026-06-05 18:07:02.27 GMT-4] [CPU: 11.38 GB] [Avail: 310.23 GB] Done in 20.415s.
> [2026-06-05 18:07:02.28 GMT-4] [CPU: 11.38 GB] [Avail: 310.23 GB] Outputting files..
> [2026-06-05 18:07:18.39 GMT-4] [CPU: 11.26 GB] [Avail: 309.58 GB] Done in 16.108s.
> [2026-06-05 18:07:18.39 GMT-4] [CPU: 11.26 GB] [Avail: 309.58 GB] Done iteration 3 in 259.627s. Total time so far 929.783s
> [2026-06-05 18:07:19.19 GMT-4] [CPU: 11.26 GB] [Avail: 309.30 GB] ----- Start Iteration 4
> [2026-06-05 18:07:19.19 GMT-4] [CPU: 11.26 GB] [Avail: 309.30 GB] Using Max Alignment Radius 64.746 (4.102A)
> [2026-06-05 18:07:19.19 GMT-4] [CPU: 11.26 GB] [Avail: 309.30 GB] Auto batchsize: 32724 in each split
> [2026-06-05 18:07:19.20 GMT-4] [CPU: 11.26 GB] [Avail: 309.30 GB] Using dynamic mask.
> [2026-06-05 18:07:50.43 GMT-4] [CPU: 11.26 GB] [Avail: 307.26 GB] -- THR 1 BATCH 500 NUM 8362 TOTAL 10.095331 ELAPSED 117.45269
--
> [2026-06-05 18:09:50.02 GMT-4] [CPU: 12.02 GB] [Avail: 301.53 GB] Processed 65008.000 images in 119.590s.
> [2026-06-05 18:09:53.28 GMT-4] [CPU: 12.02 GB] [Avail: 301.66 GB] Computing FSCs...
> [2026-06-05 18:09:53.28 GMT-4] [CPU: 12.02 GB] [Avail: 301.66 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-05 18:09:53.29 GMT-4] [CPU: 12.02 GB] [Avail: 301.66 GB] Computing FFTs on GPU.
> [2026-06-05 18:09:55.16 GMT-4] [CPU: 12.02 GB] [Avail: 300.32 GB] Done in 1.874s
> [2026-06-05 18:09:55.16 GMT-4] [CPU: 12.02 GB] [Avail: 300.32 GB] Computing cFSCs...
> [2026-06-05 18:09:57.39 GMT-4] [CPU: 12.02 GB] [Avail: 300.73 GB] Done in 2.225s
> [2026-06-05 18:09:57.39 GMT-4] [CPU: 12.02 GB] [Avail: 300.72 GB] Using Filter Radius 67.312 (3.946A) | Previous: 64.746
(4.102A)
> [2026-06-05 18:10:03.97 GMT-4] [CPU: 11.26 GB] [Avail: 301.98 GB] Non-uniform regularization with compute option: GPU
> [2026-06-05 18:10:03.98 GMT-4] [CPU: 11.26 GB] [Avail: 301.98 GB] Running local cross validation for A ...
> [2026-06-05 18:10:26.10 GMT-4] [CPU: 11.52 GB] [Avail: 303.25 GB] Local cross validation A done in 22.117s
> [2026-06-05 18:10:27.48 GMT-4] FSC Filtered Side A
```



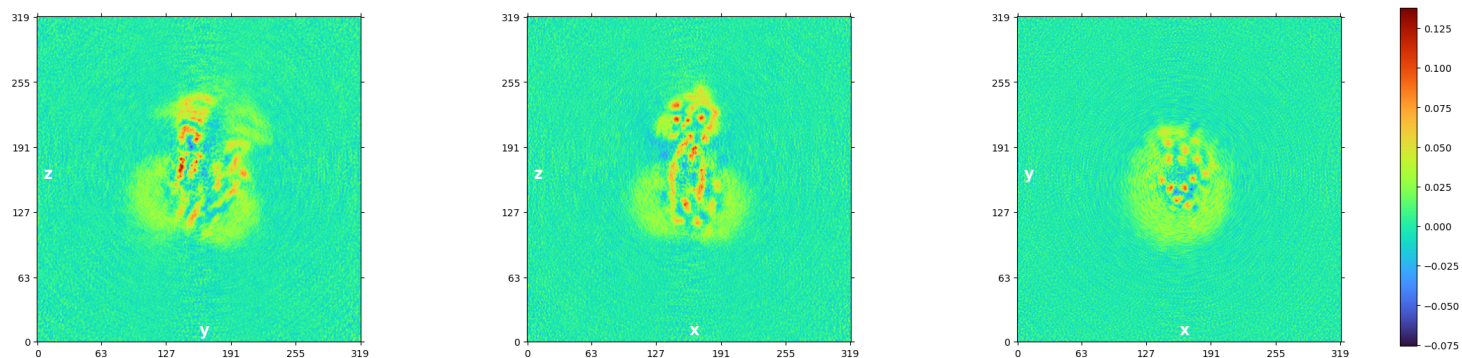
> [2026-06-05 18:10:28.67 GMT-4] CV Filtered Side A



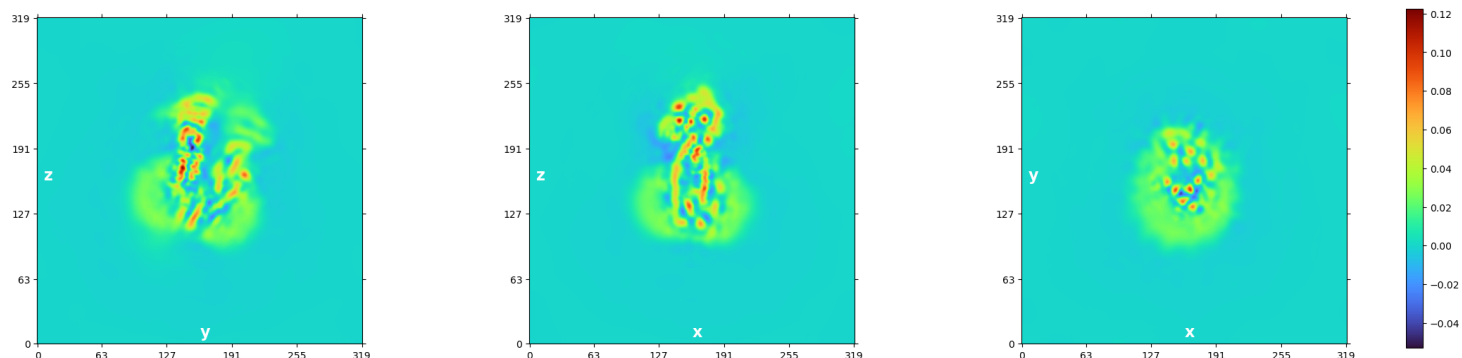
```
> [2026-06-05 18:10:28.67 GMT-4] [CPU: 11.52 GB] [Avail: 302.00 GB] Running local cross validation for B ...
```

> [2026-06-05 18:10:50.50 GMT-4] [CPU: 11.77 GB] [Avail: 305.69 GB] Local cross validation B done in 21.824s

> [2026-06-05 18:10:51.92 GMT-4] FSC Filtered Side B



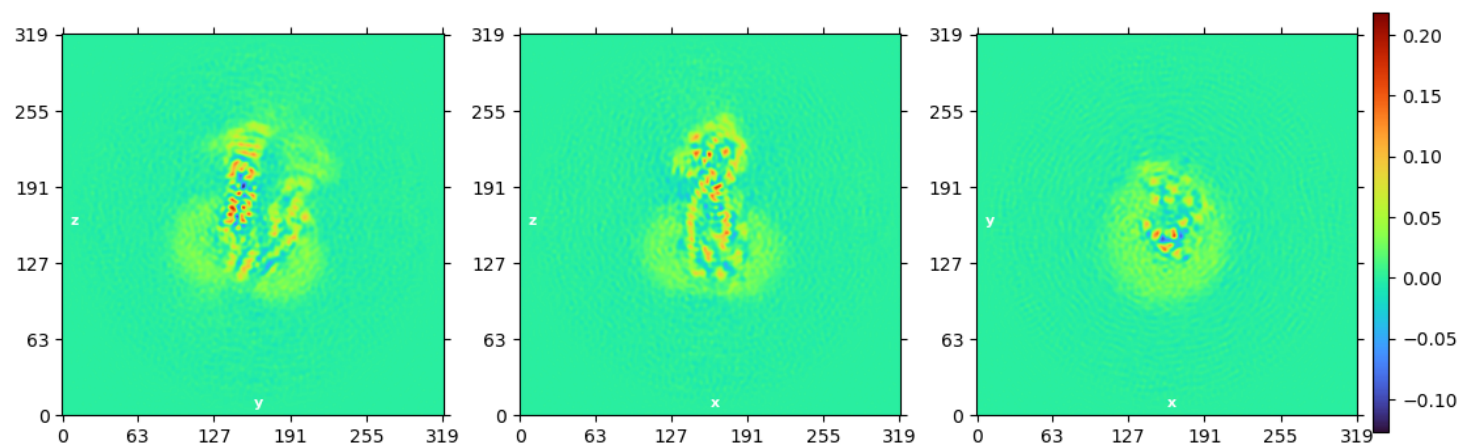
> [2026-06-05 18:10:53.03 GMT-4] CV Filtered Side B



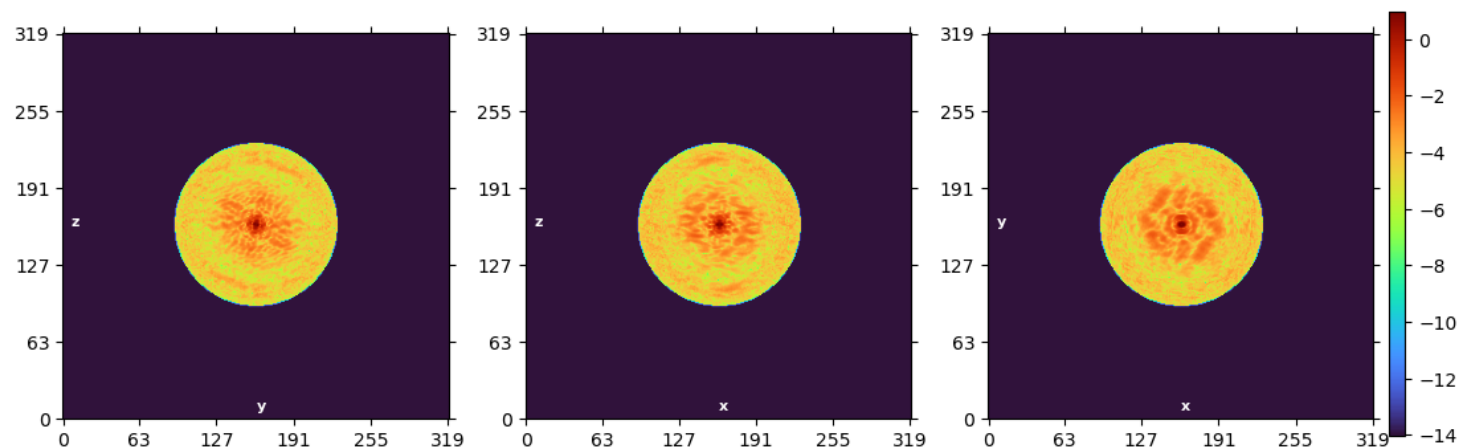
> [2026-06-05 18:11:03.24 GMT-4] [CPU: 11.79 GB] [Avail: 303.94 GB] Estimated Bfactor: -138.1

> [2026-06-05 18:11:03.25 GMT-4] [CPU: 11.79 GB] [Avail: 303.94 GB] Plotting..

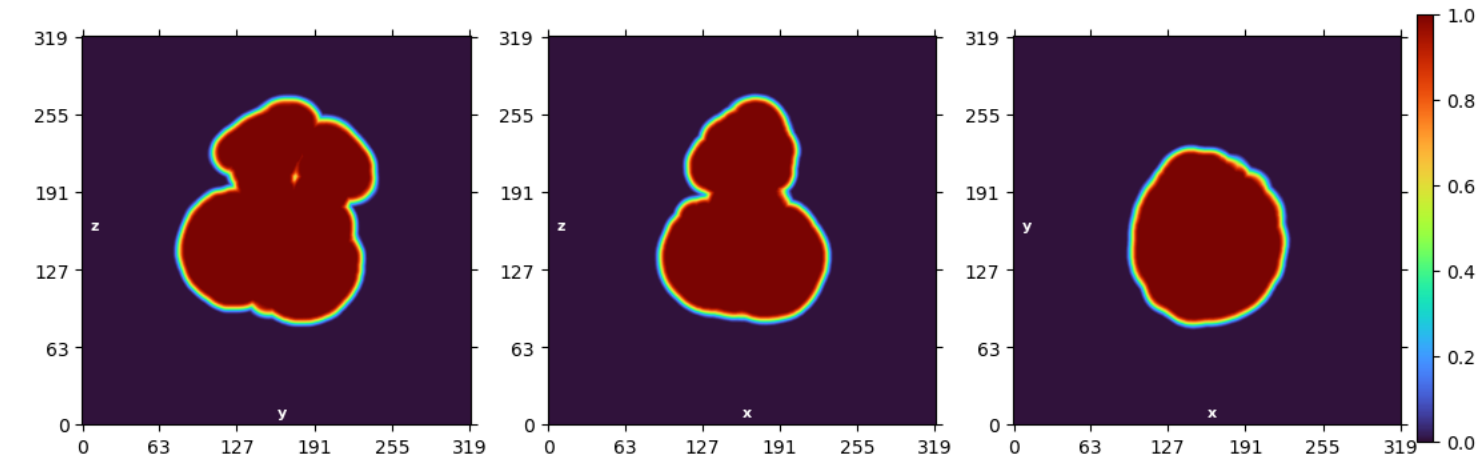
> [2026-06-05 18:11:09.92 GMT-4] Real Space Slices Iteration 004



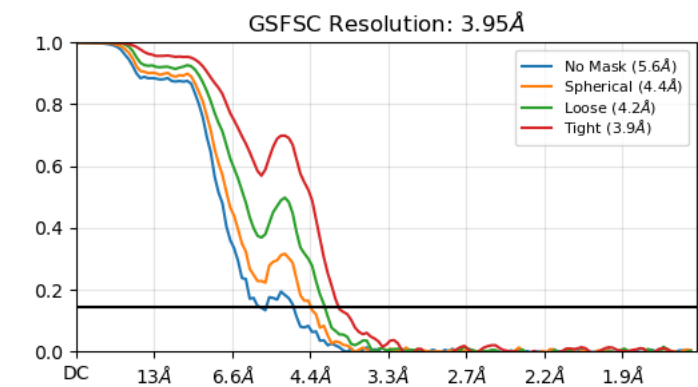
> [2026-06-05 18:11:11.32 GMT-4] Fourier Space Slices Iteration 004



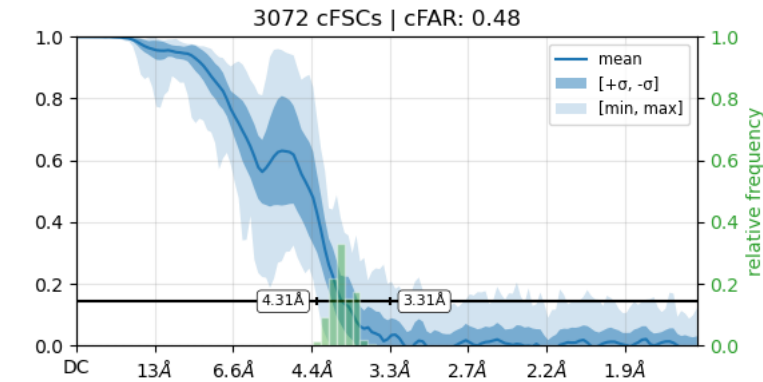
> [2026-06-05 18:11:13.38 GMT-4] Real Space Mask Slices Iteration 004



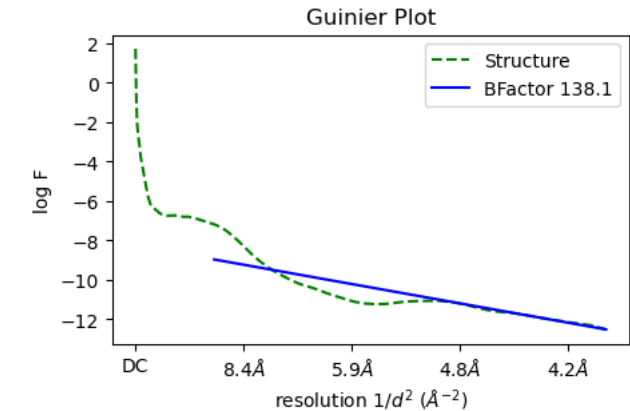
> [2026-06-05 18:11:13.89 GMT-4] FSC Iteration 004



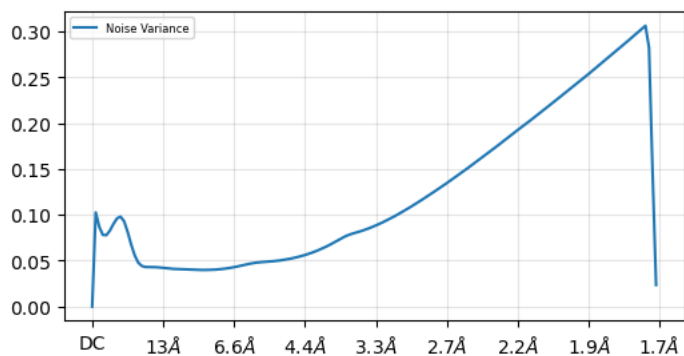
> [2026-06-05 18:11:14.62 GMT-4] cFSCs (Half-angle: 20°) Iteration 004, with tight mask



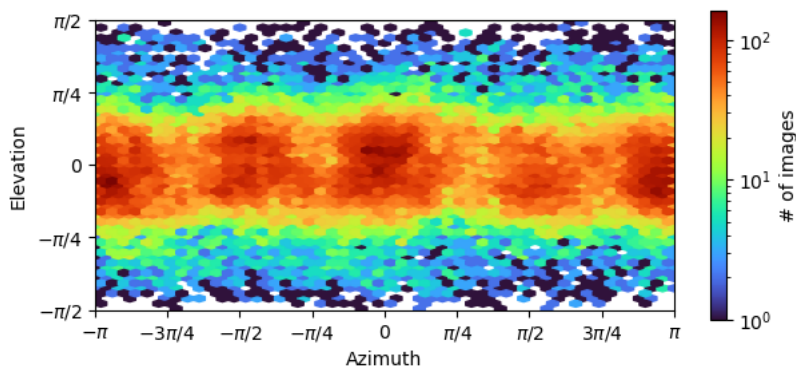
> [2026-06-05 18:11:20.69 GMT-4] Guinier Plot Iteration 004



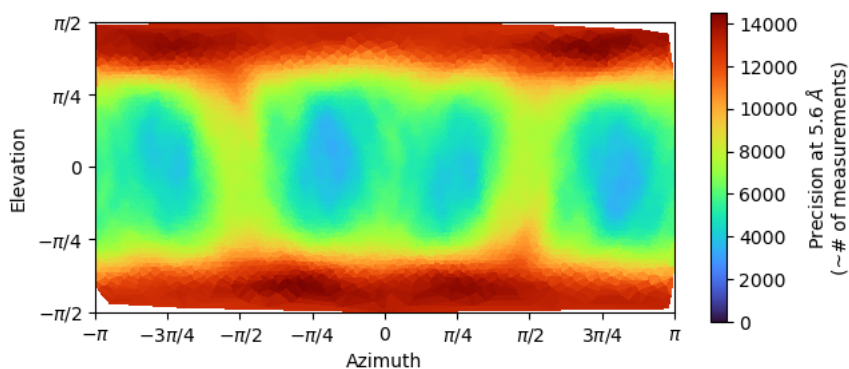
> [2026-06-05 18:11:21.08 GMT-4] Noise Model Iteration 004



> [2026-06-05 18:11:22.35 GMT-4] Viewing Direction Distribution Iteration 004

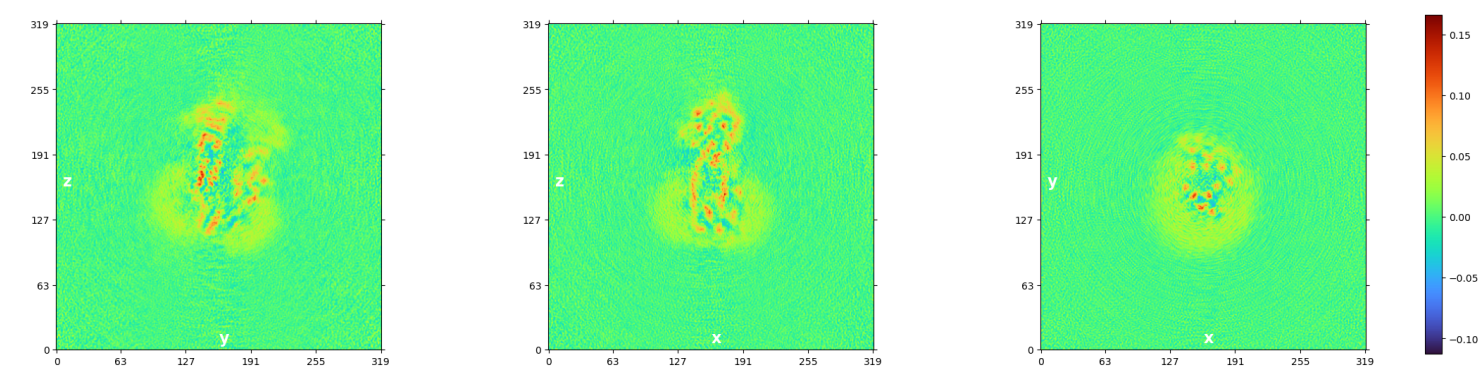


> [2026-06-05 18:11:23.96 GMT-4] Posterior Precision Directional Distribution Iteration 004

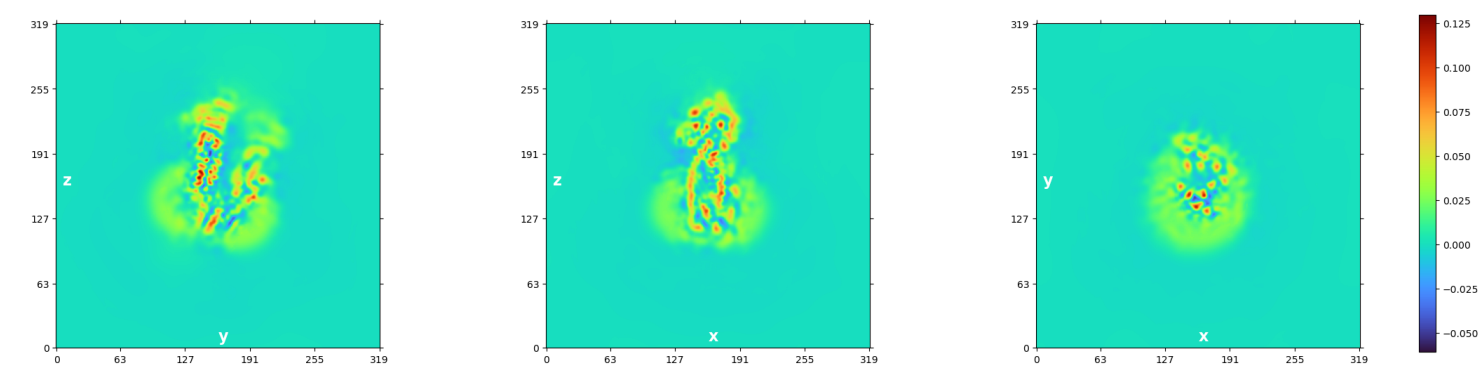


```
> [2026-06-05 18:11:23.96 GMT-4] [CPU: 11.79 GB] [Avail: 306.63 GB] Done in 20.710s.
> [2026-06-05 18:11:23.96 GMT-4] [CPU: 11.79 GB] [Avail: 306.63 GB] Outputting files..
> [2026-06-05 18:11:39.88 GMT-4] [CPU: 11.67 GB] [Avail: 311.44 GB] Done in 15.909s.
> [2026-06-05 18:11:39.88 GMT-4] [CPU: 11.67 GB] [Avail: 311.44 GB] Done iteration 4 in 260.689s. Total time so far 1191.274s
> [2026-06-05 18:11:40.59 GMT-4] [CPU: 11.67 GB] [Avail: 311.41 GB] ----- Start Iteration 5
> [2026-06-05 18:11:40.59 GMT-4] [CPU: 11.67 GB] [Avail: 311.41 GB] Using Max Alignment Radius 67.312 (3.946A)
> [2026-06-05 18:11:40.59 GMT-4] [CPU: 11.67 GB] [Avail: 311.41 GB] Using Full Dataset (split 32284 in A, 32724 in B)
> [2026-06-05 18:11:40.60 GMT-4] [CPU: 11.67 GB] [Avail: 311.41 GB] Using dynamic mask.
> [2026-06-05 18:12:09.79 GMT-4] [CPU: 11.67 GB] [Avail: 310.29 GB] -- THR 1 BATCH 500 NUM 8362 TOTAL 10.587392 ELAPSED 116.52493
--
> [2026-06-05 18:14:08.64 GMT-4] [CPU: 12.43 GB] [Avail: 312.02 GB] Processed 65008.000 images in 118.854s.
> [2026-06-05 18:14:11.79 GMT-4] [CPU: 12.46 GB] [Avail: 311.89 GB] Computing FSCs...
> [2026-06-05 18:14:11.79 GMT-4] [CPU: 12.46 GB] [Avail: 311.88 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-05 18:14:11.80 GMT-4] [CPU: 12.46 GB] [Avail: 311.88 GB] Computing FFTs on GPU.
> [2026-06-05 18:14:15.08 GMT-4] [CPU: 12.46 GB] [Avail: 311.78 GB] Done in 3.283s
> [2026-06-05 18:14:15.08 GMT-4] [CPU: 12.46 GB] [Avail: 311.78 GB] Computing cFSCs...
> [2026-06-05 18:14:17.15 GMT-4] [CPU: 12.46 GB] [Avail: 311.71 GB] Done in 2.066s
> [2026-06-05 18:14:17.16 GMT-4] [CPU: 12.46 GB] [Avail: 311.71 GB] Using Filter Radius 70.409 (3.772A) | Previous: 67.312
(3.946A)
> [2026-06-05 18:14:23.15 GMT-4] [CPU: 11.70 GB] [Avail: 312.27 GB] Non-uniform regularization with compute option: GPU
> [2026-06-05 18:14:23.15 GMT-4] [CPU: 11.70 GB] [Avail: 312.27 GB] Running local cross validation for A ...
> [2026-06-05 18:14:45.83 GMT-4] [CPU: 11.92 GB] [Avail: 313.37 GB] Local cross validation A done in 22.671s
```

> [2026-06-05 18:14:47.06 GMT-4] FSC Filtered Side A



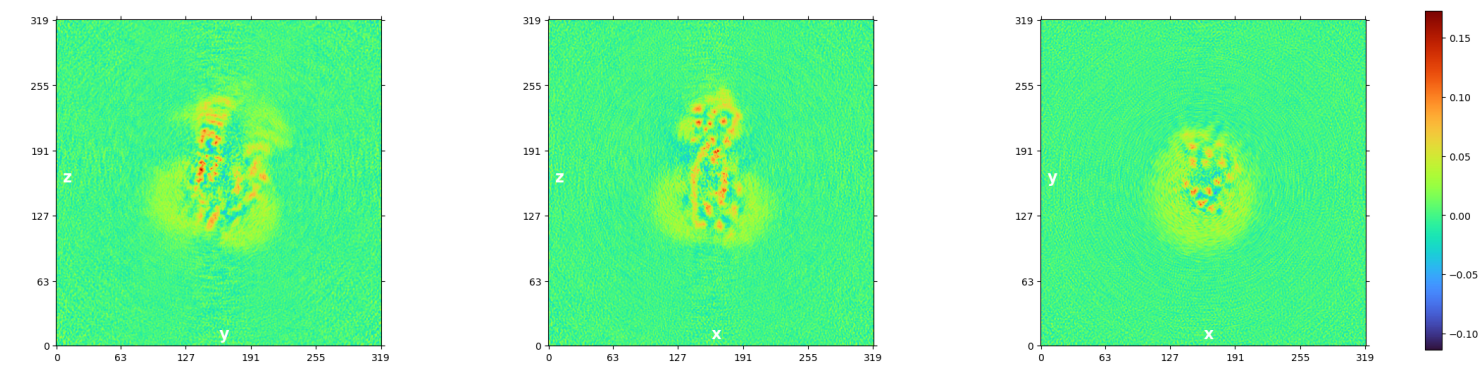
> [2026-06-05 18:14:47.97 GMT-4] CV Filtered Side A



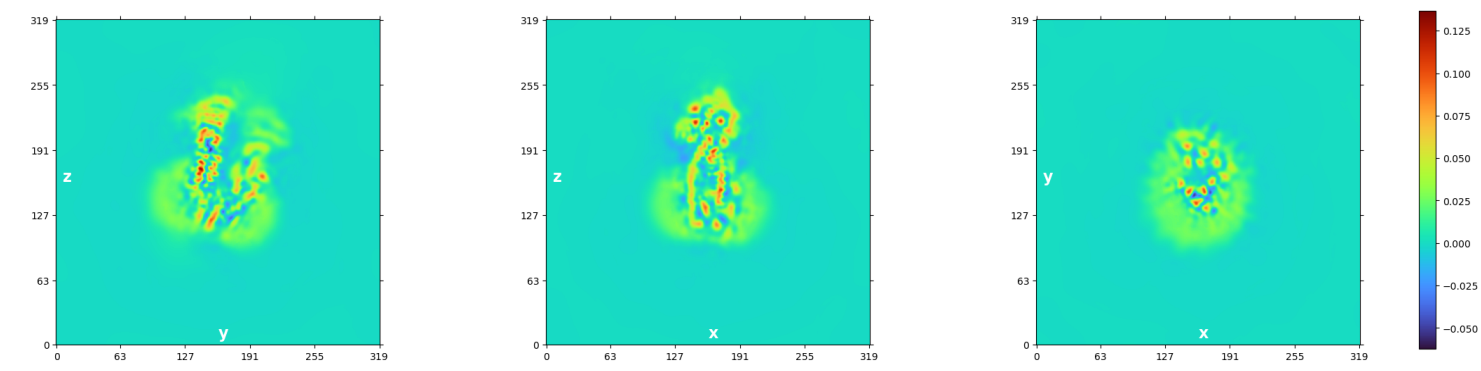
> [2026-06-05 18:14:47.98 GMT-4] [CPU: 11.92 GB] [Avail: 313.30 GB] Running local cross validation for B ...

> [2026-06-05 18:15:09.79 GMT-4] [CPU: 12.17 GB] [Avail: 312.37 GB] Local cross validation B done in 21.804s

> [2026-06-05 18:15:11.43 GMT-4] FSC Filtered Side B



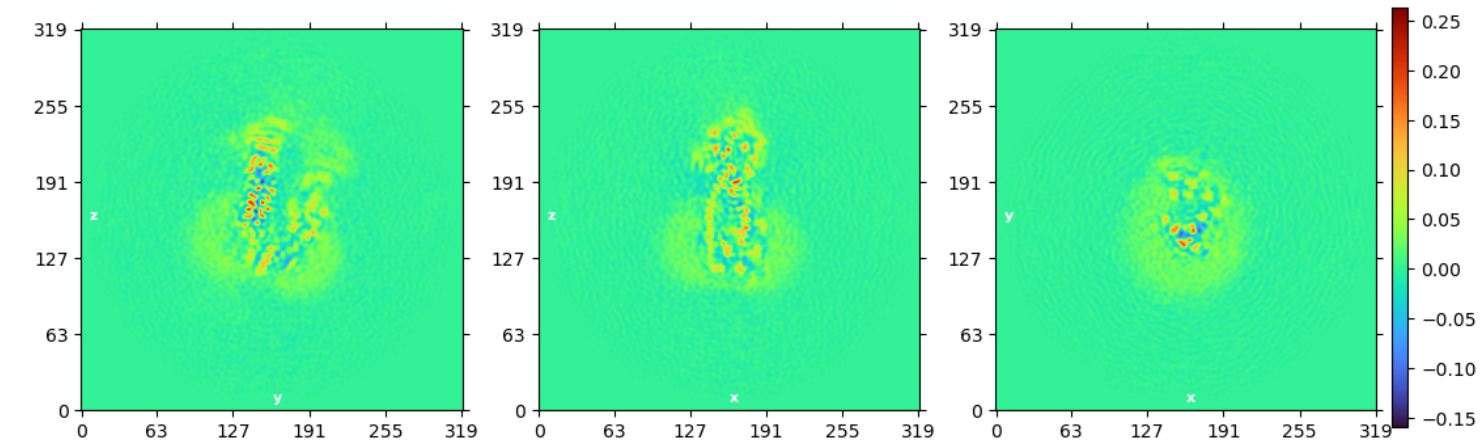
> [2026-06-05 18:15:12.45 GMT-4] CV Filtered Side B



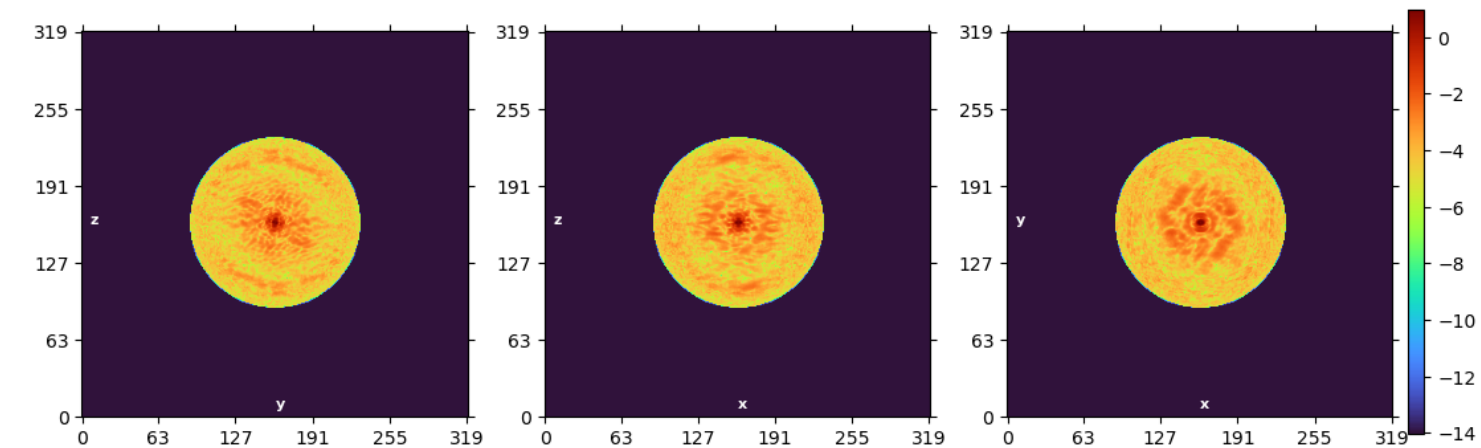
> [2026-06-05 18:15:22.81 GMT-4] [CPU: 12.23 GB] [Avail: 311.73 GB] Estimated Bfactor: -126.4

> [2026-06-05 18:15:22.82 GMT-4] [CPU: 12.23 GB] [Avail: 311.73 GB] Plotting..

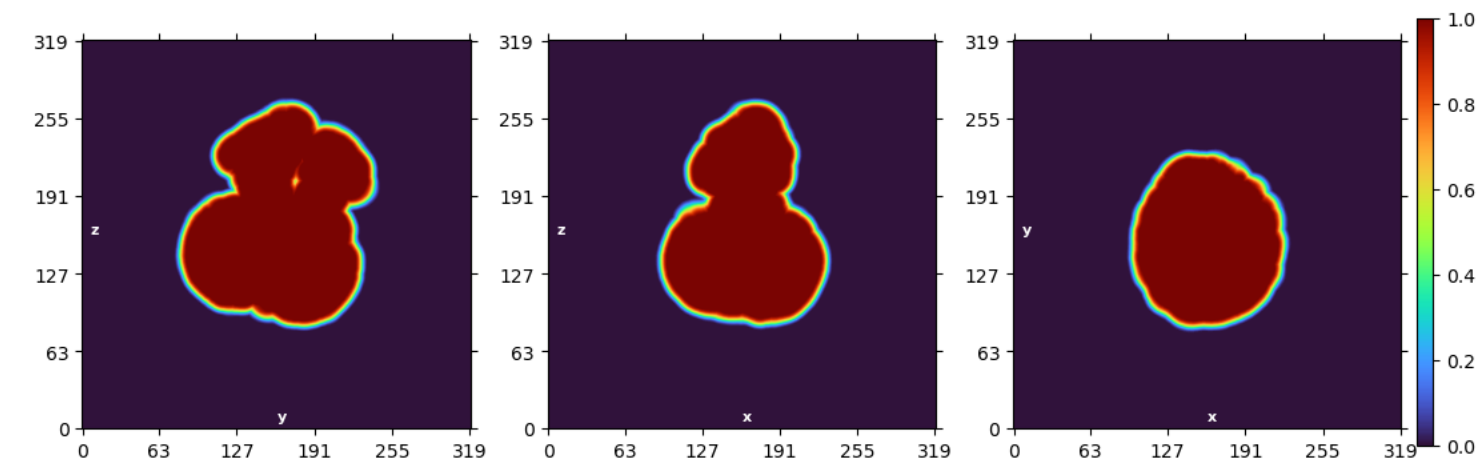
> [2026-06-05 18:15:29.62 GMT-4] Real Space Slices Iteration 005



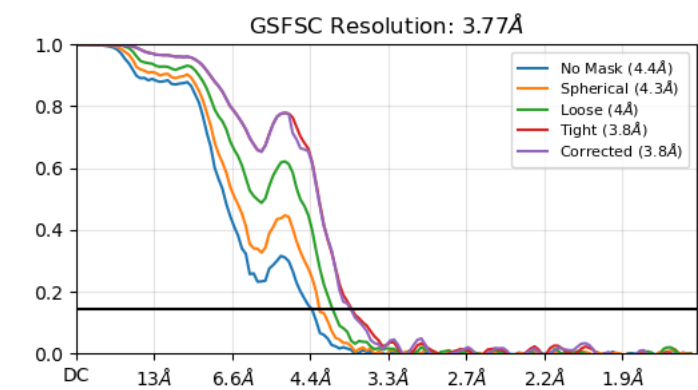
> [2026-06-05 18:15:31.05 GMT-4] Fourier Space Slices Iteration 005



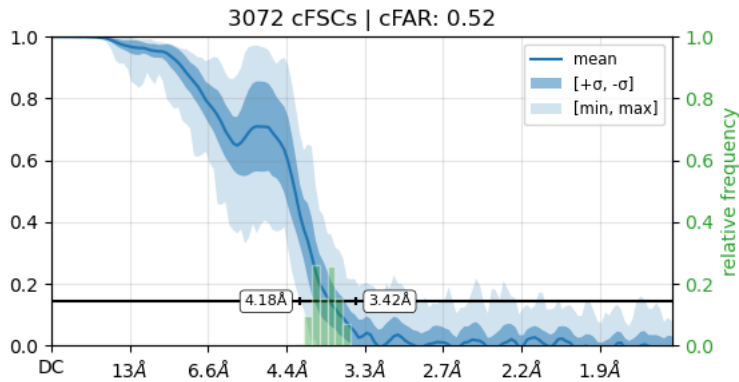
> [2026-06-05 18:15:33.18 GMT-4] Real Space Mask Slices Iteration 005



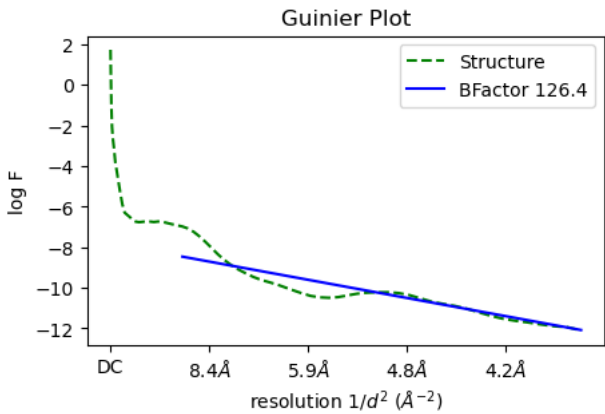
> [2026-06-05 18:15:33.73 GMT-4] FSC Iteration 005



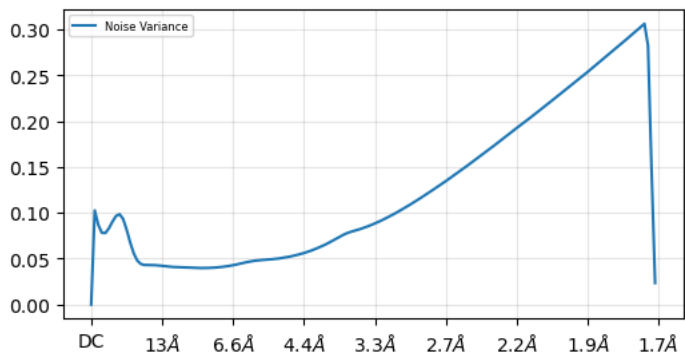
> [2026-06-05 18:15:34.50 GMT-4] cFSCs (Half-angle: 20°) Iteration 005, with tight mask



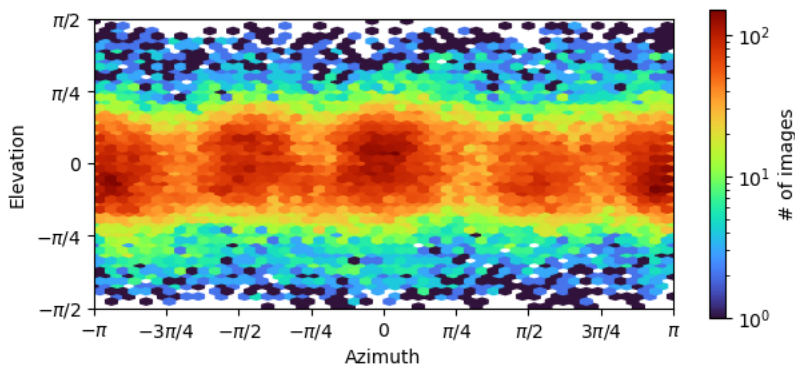
> [2026-06-05 18:15:40.56 GMT-4] Guinier Plot Iteration 005



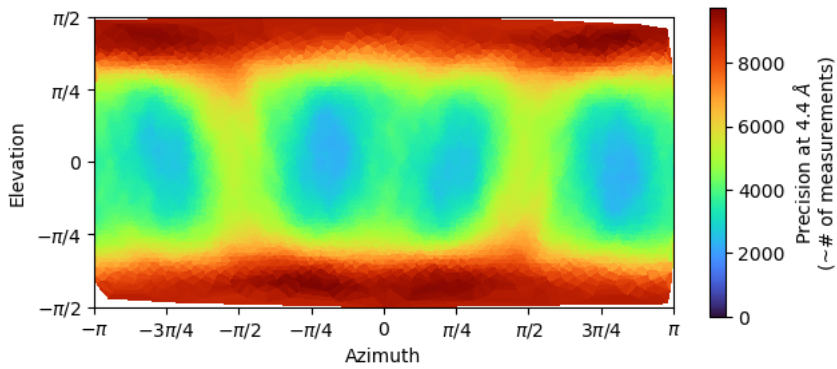
> [2026-06-05 18:15:40.94 GMT-4] Noise Model Iteration 005



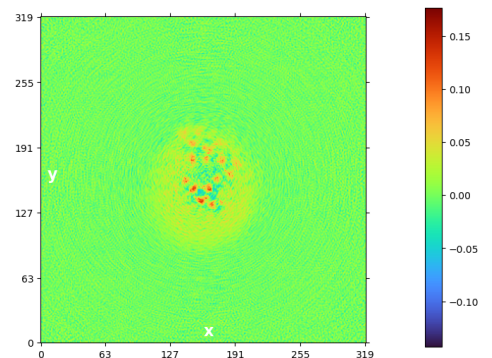
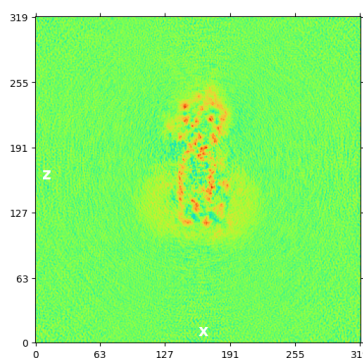
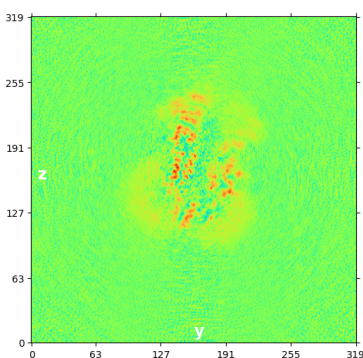
> [2026-06-05 18:15:42.15 GMT-4] Viewing Direction Distribution Iteration 005



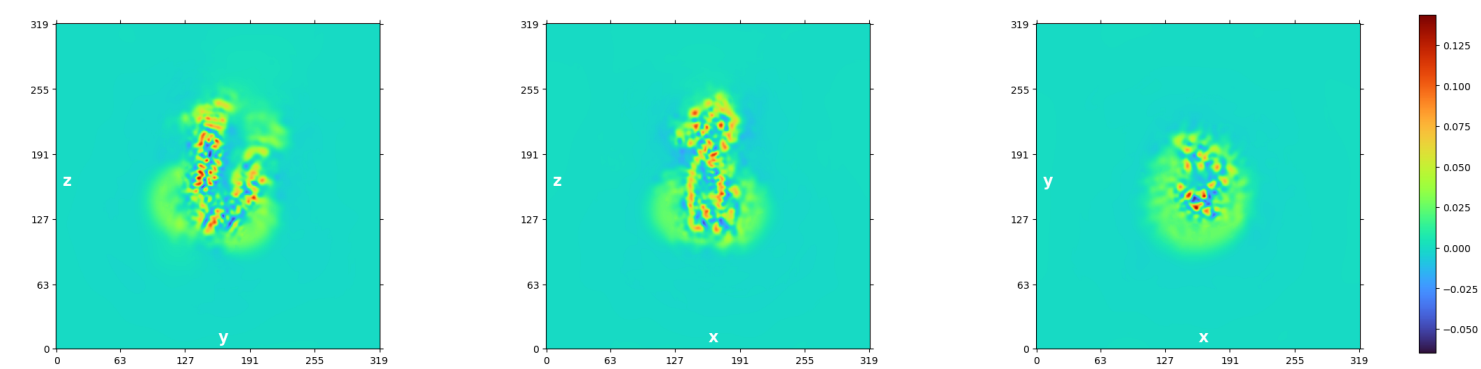
> [2026-06-05 18:15:43.77 GMT-4] Posterior Precision Directional Distribution Iteration 005



```
> [2026-06-05 18:15:43.77 GMT-4] [CPU: 12.23 GB] [Avail: 309.98 GB] Done in 20.942s.
> [2026-06-05 18:15:43.77 GMT-4] [CPU: 12.23 GB] [Avail: 309.98 GB] Outputting files..
> [2026-06-05 18:15:59.88 GMT-4] [CPU: 12.11 GB] [Avail: 307.21 GB] Done in 16.109s.
> [2026-06-05 18:15:59.89 GMT-4] [CPU: 12.11 GB] [Avail: 307.21 GB] Done iteration 5 in 259.296s. Total time so far 1451.282s
> [2026-06-05 18:16:00.87 GMT-4] [CPU: 12.11 GB] [Avail: 307.17 GB] ----- Start Iteration 6
> [2026-06-05 18:16:00.87 GMT-4] [CPU: 12.11 GB] [Avail: 307.17 GB] Using Max Alignment Radius 70.409 (3.772A)
> [2026-06-05 18:16:00.88 GMT-4] [CPU: 12.11 GB] [Avail: 307.17 GB] Using Full Dataset (split 32284 in A, 32724 in B)
> [2026-06-05 18:16:00.88 GMT-4] [CPU: 12.11 GB] [Avail: 307.17 GB] Using dynamic mask.
> [2026-06-05 18:16:31.92 GMT-4] [CPU: 12.11 GB] [Avail: 307.11 GB] -- THR 1 BATCH 500 NUM 8362 TOTAL 10.495171 ELAPSED 116.73662
--
> [2026-06-05 18:18:30.88 GMT-4] [CPU: 12.87 GB] [Avail: 303.32 GB] Processed 65008.000 images in 118.962s.
> [2026-06-05 18:18:34.31 GMT-4] [CPU: 12.90 GB] [Avail: 304.12 GB] Computing FSCs...
> [2026-06-05 18:18:34.32 GMT-4] [CPU: 12.90 GB] [Avail: 304.12 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-05 18:18:34.32 GMT-4] [CPU: 12.90 GB] [Avail: 304.12 GB] Computing FFTs on GPU.
> [2026-06-05 18:18:36.36 GMT-4] [CPU: 12.90 GB] [Avail: 304.17 GB] Done in 2.042s
> [2026-06-05 18:18:36.38 GMT-4] [CPU: 12.90 GB] [Avail: 304.17 GB] Optimizing FSC Mask...
> [2026-06-05 18:18:36.42 GMT-4] [CPU: 12.90 GB] [Avail: 304.16 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-05 18:18:36.42 GMT-4] [CPU: 12.90 GB] [Avail: 304.16 GB] Computing FFTs on GPU.
> [2026-06-05 18:18:38.52 GMT-4] [CPU: 13.15 GB] [Avail: 303.60 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-05 18:18:38.52 GMT-4] [CPU: 13.15 GB] [Avail: 303.60 GB] Computing FFTs on GPU.
> [2026-06-05 18:18:40.56 GMT-4] [CPU: 13.15 GB] [Avail: 304.12 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-05 18:18:40.57 GMT-4] [CPU: 13.15 GB] [Avail: 304.12 GB] Computing FFTs on GPU.
> [2026-06-05 18:18:42.68 GMT-4] [CPU: 13.15 GB] [Avail: 304.71 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-05 18:18:42.68 GMT-4] [CPU: 13.15 GB] [Avail: 304.71 GB] Computing FFTs on GPU.
> [2026-06-05 18:18:44.80 GMT-4] [CPU: 13.15 GB] [Avail: 303.78 GB] Done in 8.412s
> [2026-06-05 18:18:44.80 GMT-4] [CPU: 13.15 GB] [Avail: 303.78 GB] Computing cFSCs...
> [2026-06-05 18:18:46.94 GMT-4] [CPU: 13.15 GB] [Avail: 304.78 GB] Done in 2.139s
> [2026-06-05 18:18:46.95 GMT-4] [CPU: 13.15 GB] [Avail: 304.77 GB] Using Filter Radius 73.467 (3.615A) | Previous: 70.409
(3.772A)
> [2026-06-05 18:18:53.59 GMT-4] [CPU: 12.39 GB] [Avail: 305.58 GB] Non-uniform regularization with compute option: GPU
> [2026-06-05 18:18:53.59 GMT-4] [CPU: 12.39 GB] [Avail: 305.58 GB] Running local cross validation for A ...
> [2026-06-05 18:19:16.37 GMT-4] [CPU: 12.62 GB] [Avail: 306.08 GB] Local cross validation A done in 22.776s
> [2026-06-05 18:19:17.90 GMT-4] FSC Filtered Side A
```



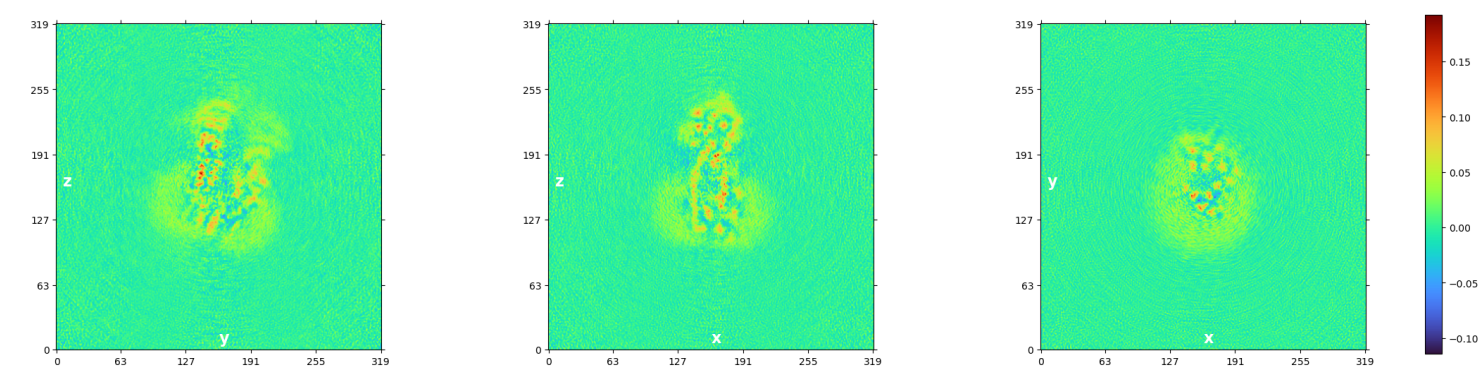
> [2026-06-05 18:19:19.14 GMT-4] CV Filtered Side A



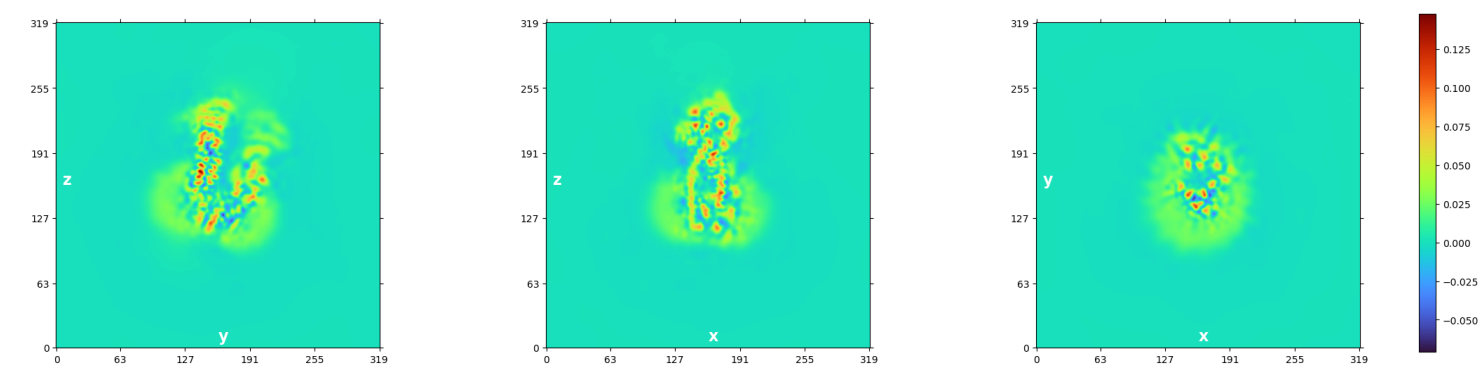
> [2026-06-05 18:19:19.14 GMT-4] [CPU: 12.62 GB] [Avail: 305.03 GB] Running local cross validation for B ...

> [2026-06-05 18:19:40.73 GMT-4] [CPU: 12.87 GB] [Avail: 306.92 GB] Local cross validation B done in 21.584s

> [2026-06-05 18:19:42.11 GMT-4] FSC Filtered Side B



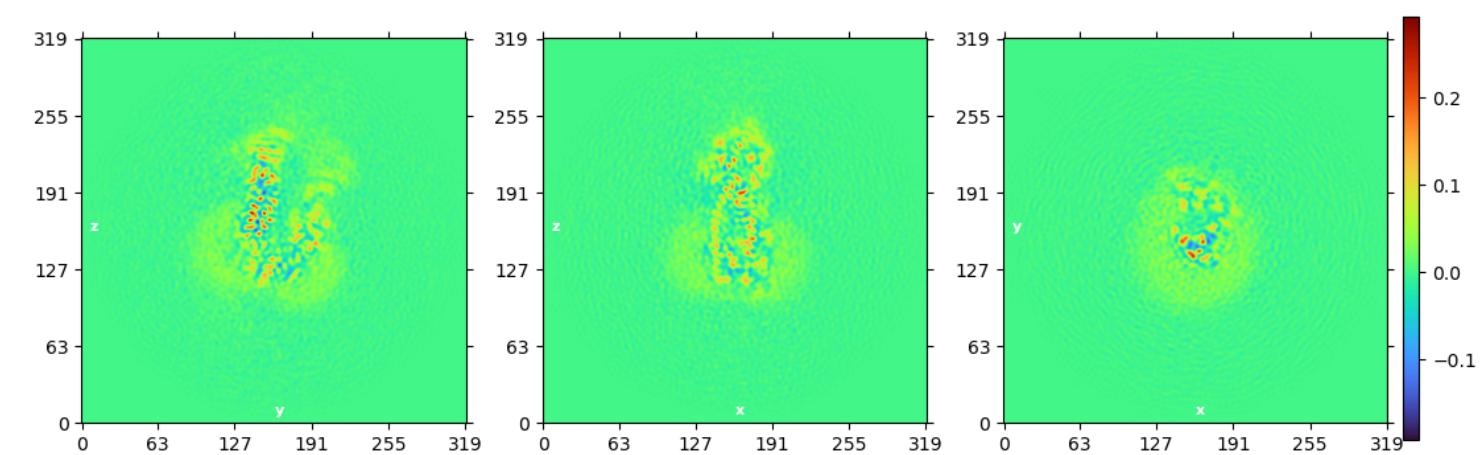
> [2026-06-05 18:19:43.14 GMT-4] CV Filtered Side B



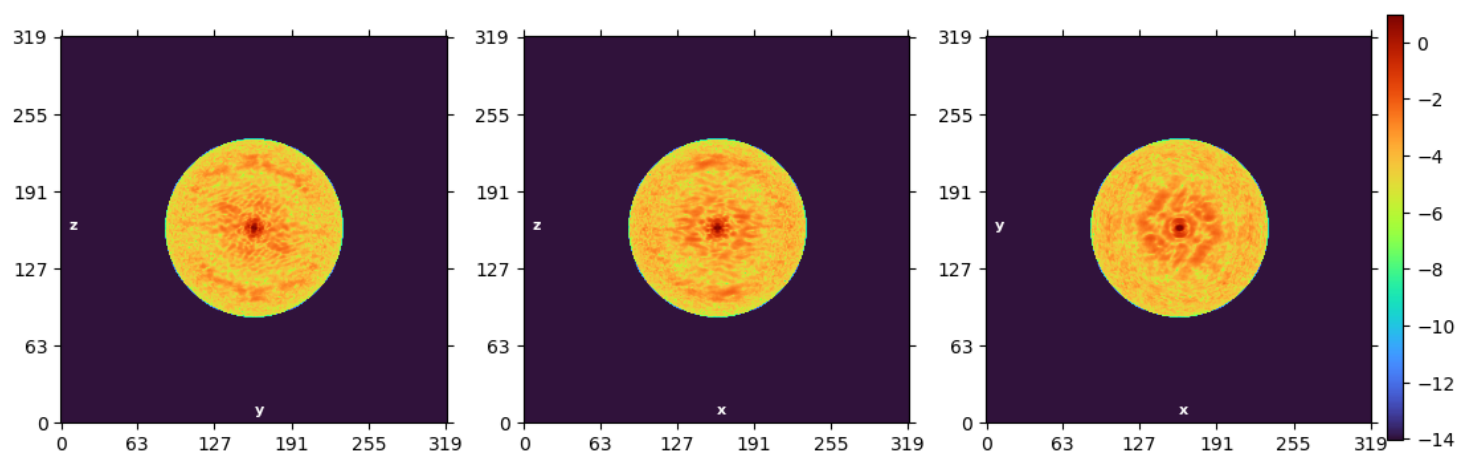
> [2026-06-05 18:19:53.85 GMT-4] [CPU: 12.89 GB] [Avail: 304.59 GB] Estimated Bfactor: -117.5

> [2026-06-05 18:19:53.86 GMT-4] [CPU: 12.89 GB] [Avail: 304.59 GB] Plotting..

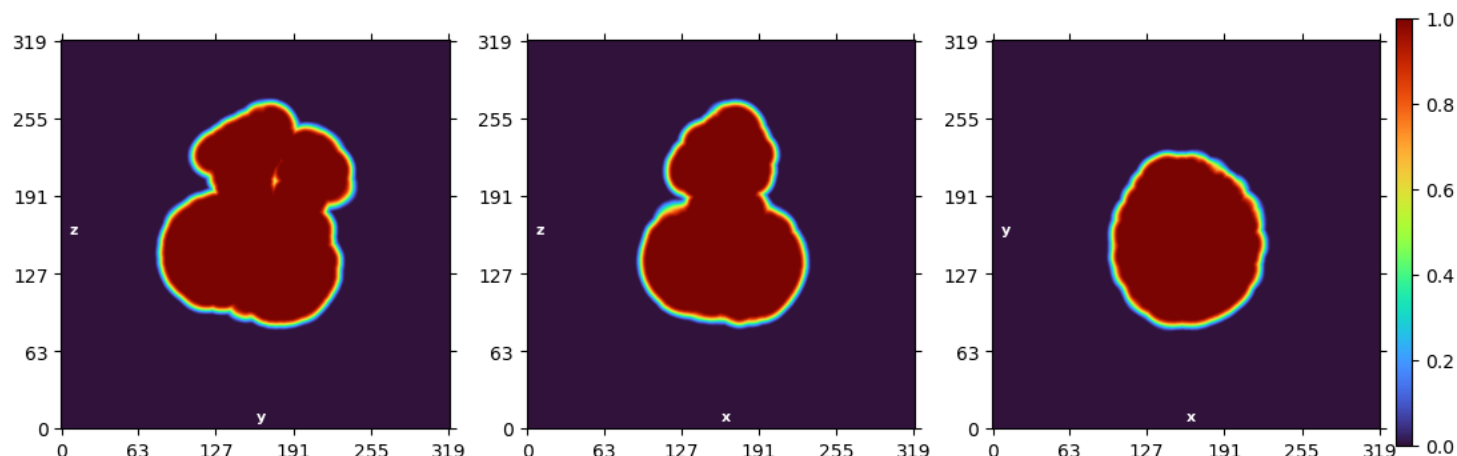
> [2026-06-05 18:19:59.81 GMT-4] Real Space Slices Iteration 006



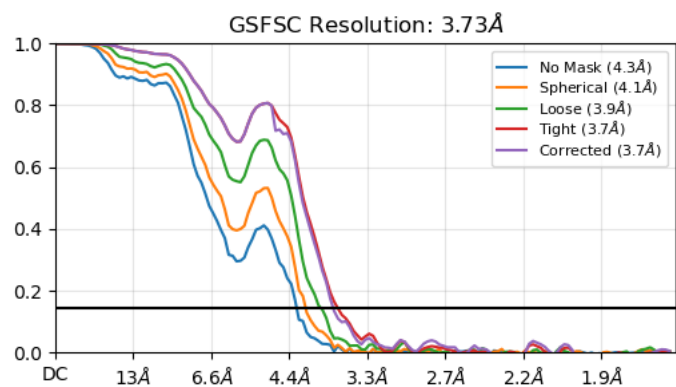
> [2026-06-05 18:20:01.36 GMT-4] Fourier Space Slices Iteration 006



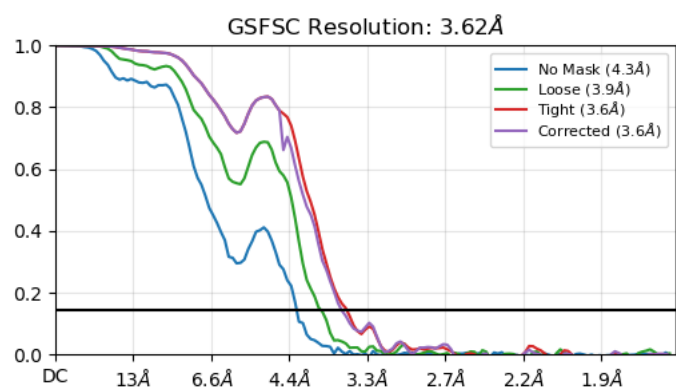
> [2026-06-05 18:20:03.41 GMT-4] Real Space Mask Slices Iteration 006



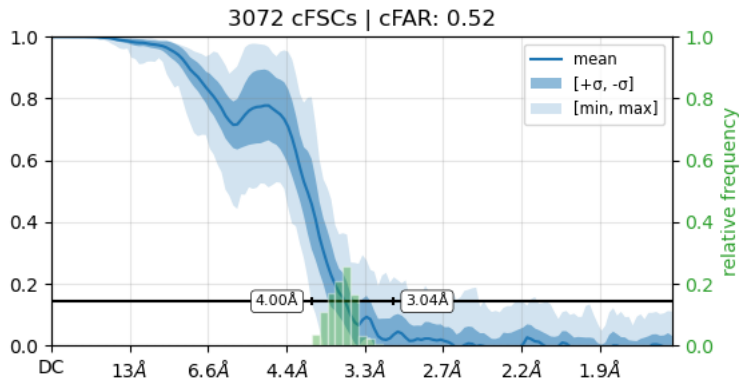
> [2026-06-05 18:20:04.07 GMT-4] FSC Iteration 006



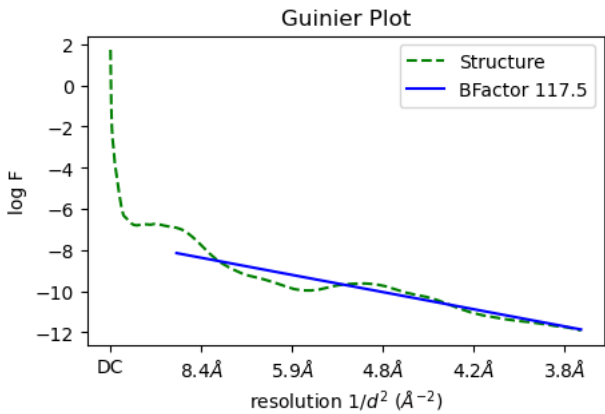
> [2026-06-05 18:20:04.59 GMT-4] FSC Iteration 006, after FSC-mask auto-tightening



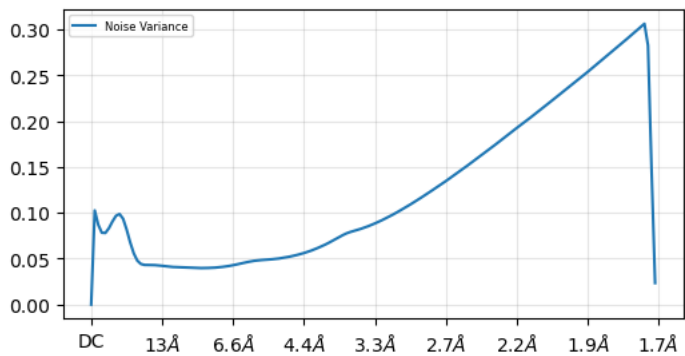
> [2026-06-05 18:20:05.30 GMT-4] cFSCs (Half-angle: 20°) Iteration 006, with autotight mask



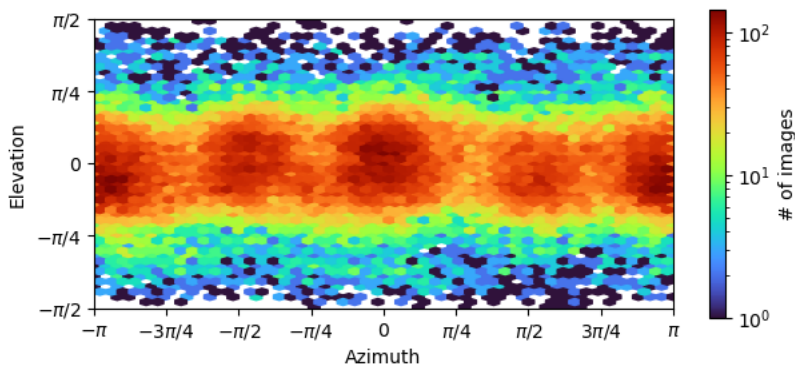
> [2026-06-05 18:20:10.80 GMT-4] Guinier Plot Iteration 006



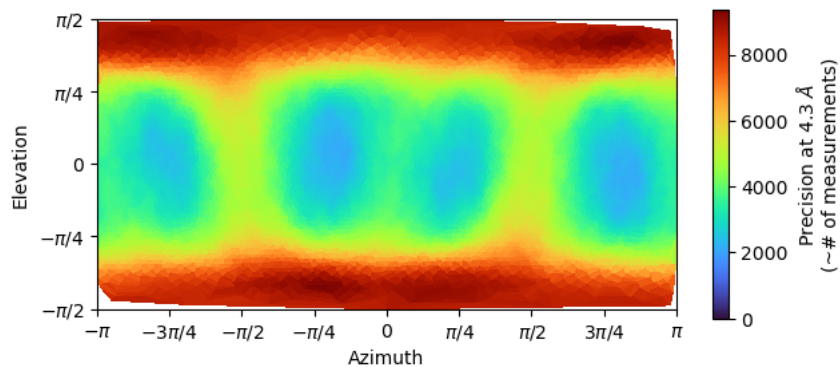
> [2026-06-05 18:20:11.21 GMT-4] Noise Model Iteration 006



> [2026-06-05 18:20:12.59 GMT-4] Viewing Direction Distribution Iteration 006



> [2026-06-05 18:20:14.24 GMT-4] Posterior Precision Directional Distribution Iteration 006



```
> [2026-06-05 18:20:14.24 GMT-4] [CPU: 12.90 GB] [Avail: 305.38 GB] Done in 20.384s.
> [2026-06-05 18:20:14.25 GMT-4] [CPU: 12.90 GB] [Avail: 305.38 GB] Outputting files..
> [2026-06-05 18:20:31.35 GMT-4] [CPU: 12.80 GB] [Avail: 307.00 GB] Done in 17.101s.
> [2026-06-05 18:20:31.36 GMT-4] [CPU: 12.80 GB] [Avail: 307.00 GB] Done iteration 6 in 270.480s. Total time so far 1722.749s
> [2026-06-05 18:20:32.10 GMT-4] [CPU: 12.80 GB] [Avail: 306.89 GB] ===== Done Refinement =====
> [2026-06-05 18:20:32.10 GMT-4] [CPU: 12.80 GB] [Avail: 306.89 GB] Note that the output structure from refinement has been
                                                                    lowpass filtered to the gold-standard FSC resolution
                                                                    estimated
                                                                    above. However, the structure probably needs sharpening in
                                                                    order to best visualize high resolution features. This can be
                                                                    done with the Sharpening task (as a new experiment).

> [2026-06-05 18:20:32.10 GMT-4] [CPU: 12.80 GB] [Avail: 306.89 GB] Full run took 1723.499s
> [2026-06-05 18:20:32.71 GMT-4] [CPU: 6.94 GB] [Avail: 313.34 GB] -----
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> [2026-06-05 18:20:32.72 GMT-4] [CPU: 6.94 GB] [Avail: 313.35 GB] Compiling job outputs...
> [2026-06-05 18:20:32.72 GMT-4] [CPU: 6.94 GB] [Avail: 313.35 GB] Passing through outputs for output group particles from input
                                                                    group particles
> [2026-06-05 18:20:32.90 GMT-4] [CPU: 6.94 GB] [Avail: 313.34 GB] This job outputted results ['alignments3D', 'ctf']
> [2026-06-05 18:20:32.90 GMT-4] [CPU: 6.94 GB] [Avail: 313.33 GB] Loaded output dataset with 65008 items
> [2026-06-05 18:20:32.91 GMT-4] [CPU: 6.94 GB] [Avail: 313.33 GB] Passthrough results ['blob']
> [2026-06-05 18:20:33.21 GMT-4] [CPU: 6.95 GB] [Avail: 313.31 GB] Loaded passthrough dataset with 65008 items
> [2026-06-05 18:20:33.24 GMT-4] [CPU: 6.95 GB] [Avail: 313.31 GB] Intersection of output and passthrough has 65008 items
> [2026-06-05 18:20:33.34 GMT-4] [CPU: 6.95 GB] [Avail: 313.30 GB] Output dataset contains: ['blob']
> [2026-06-05 18:20:33.34 GMT-4] [CPU: 6.95 GB] [Avail: 313.30 GB] Outputting passthrough result blob
> [2026-06-05 18:20:33.35 GMT-4] [CPU: 6.95 GB] [Avail: 313.30 GB] Checking outputs for output group particles
> [2026-06-05 18:20:33.60 GMT-4] [CPU: 6.95 GB] [Avail: 313.28 GB] Updating job size...
> [2026-06-05 18:20:35.26 GMT-4] [CPU: 6.95 GB] [Avail: 312.99 GB] Exporting job and creating csg files...
> [2026-06-05 18:20:36.73 GMT-4] [CPU: 6.95 GB] [Avail: 312.92 GB] *****
                                                                    **

> [2026-06-05 18:20:36.74 GMT-4] [CPU: 6.95 GB] [Avail: 312.92 GB] Job complete. Total time 1728.17s
```